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WaveformDisplayBurstHRAM.3.01

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Bursting, Actual Mode, and HRAM

Bursting, Actual Mode, and HRAM

The section includes the following topics:

- [Bursting Patterns](#)

- [HRAM Setup](#)

- [Using Actual Mode](#)

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WaveformDisplayBurstHRAM.3.02

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Bursting, Actual Mode, and HRAM : Bursting Patterns

Bursting Patterns

You can reburst the most recently burst pattern or pattern set. This lets you debug the patterns. The following topics are covered:

- Burst Commands
- Burst Time Align
- Masking Pins
- Refreshing Other Digital Tools
- Turning Off Automatic Rebursting

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WaveformDisplayBurstHRAM.3.03



[Bursting, Actual Mode, and HRAM](#) : [Bursting Patterns](#) : Burst Commands

Burst Commands

Burst commands are all on the [Debug](#) menu. In addition, they are on the toolbar.

 **Burst** By default, the Burst command appears on the button.

If you click the triangle, other commands appear. Select any command. The last selected command then appears on the button.

Burst always acts on the last burst patterns or pattern sets. The Burst commands and buttons are as follows:

 Burst	Rebursts the last burst patterns or pattern sets.
 Burst Loop	Rebursts in a loop. Use Halt to stop the loop. The status bar Burst Count indicates the number of times the pattern has been burst.
 Burst Until Pass	Rebursts in a loop, until the first pass on an HSD pin is detected. The status bar Burst Count indicates the number of times the pattern has been burst. You can use Halt to stop the loop.
 Burst Until Fail	Rebursts in a loop until the first failure on an HSD pin is detected. If no failures are detected, you can use the Halt button or command to stop rebursting. The Burst Count, the number of times the pattern was burst, is displayed in the status bar.
 Burst Time Align	Rebursts a pattern with both HSD and serial pins in a way that permits correct time alignment. See Burst Time Align .

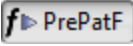
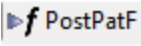
All time domains are included in a reburst.

Bursting rebursts the last burst patterns or pattern sets in the job. This could be different from the ones currently viewable in the tool.

Waveform Display automatically refreshes after a reburst. If the patterns reburst are different from the ones currently displayed, all checked pins are cleared and nothing is displayed. You need to check the items in the Burst Explorer to display data.

 While a pattern is bursting, the Halt button is enabled, letting you halt the burst.

The test instance that bursts the pattern as part of the job may specify an interpose function to be executed before the pattern burst (PrePatF) or after the pattern burst (PostPatF). When you burst these patterns in Waveform Display, it may be desirable to execute one or both of these interpose functions as well. Waveform Display lets you enable or disable the execution of one or both functions:

-  or Debug>PrePatF: Enable or disable the execution of the PrePatF function before each pattern burst.
-  or Debug>PostPatF: Enable or disable the execution of the PostPatF function after each pattern burst.

Note that these PrePatF and PostPatF settings are uniform across PatternTool, Waveform Display, and HRAM Display. Changing a setting in one tool changes it for the others as well. In this way, pattern reburst is the same across all tools.





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Bursting, Actual Mode, and HRAM : [Bursting Patterns](#) : Burst Time Align

Burst Time Align

 **Burst Time Align** The Burst Time Align command is useful when a pattern contains both HSD and serial pins:

Because of the hardware implementation, displays for the HSD and serial pins are not correctly aligned with respect to each other and to the time axis. This command provides a means of getting an alignment that is approximately correct.

The steps are as follows:

1. [Select the pins to be displayed](#), at least one HSD and one serial pin.
2. [Run Burst Time Align](#).
3. [Note the time of the first Start command](#) in on the serial pin.
4. [Set up HSD HRAM to start capture on the first Start](#). On the HRAM Setup panel, do the following:
 - a. [Check the box](#) Manually configure HRAM.
 - b. [Check the box](#) Trigger specified event.
 - c. [Under Patgen Event](#), check Time: and enter the time of the first Start.

5. [Run](#) Burst Time Align again. The HSD and serial pins should now be correctly aligned.

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Bursting, Actual Mode, and HRAM : [Bursting Patterns](#) : Masking Pins

Masking Pins

You can mask a pin before rebursting. Masking failures is a common debug activity that lets you concentrate on a subset of failures of interest.

To mask a pin, select it in the Burst Explorer. Then right-click and select Mask from the context menu. The command Debug>Mask is also enabled when you select a pin in the Burst Explorer. You can select and mask an entire pin group or list or a single pin in a group or list.

When a pin is masked, it is displayed in blue in the Burst Explorer. If you mask a pin in a pin group or list, the group or list name is also displayed in blue.

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Bursting, Actual Mode, and HRAM : [Bursting Patterns](#) : Refreshing Other Digital Tools

Refreshing Other Digital Tools

If you have other digital tools, such as [PatternTool](#), open when you burst a pattern in [Waveform Display](#), the other tools must be refreshed after the burst.

Auto Refresh Digital Tools You can have the other tools automatically refreshed by clicking this button. By default, the feature is enabled.

Other tools are refreshed only if you reburst the same pattern. If you step through the test and burst a different pattern, [Waveform Display](#) and the other tools must be manually refreshed.

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Bursting, Actual Mode, and HRAM : [Bursting Patterns](#) : Turning Off Automatic Rebursting

Turning Off Automatic Rebursting

Automatically Reburst

By default, Waveform Display automatically performs a reburst when you click Refresh or bring up the tool. You can use this button or Tools>Automatically Reburst to turn the automatic reburst on or off.

If Automatically Reburst if off, the tool displays the existing contents of HRAM. If HRAM does not contain a contiguous set of vectors, a warning message appears.

It can be useful to turn off Automatically Reburst if you want to look at a pattern with a CPU loop. In this case, rebursting the pattern can cause the tool to be stuck, because you would need to run some code in VBA to get the pattern out of the loops.

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WaveformDisplayBurstHRA3.08



Bursting, Actual Mode, and HRA : HRA Setup

HRA Setup

The HRA Setup panel lets you configure HRA for your current Waveform Display session.

If the HRA Setup panel is not visible, use View>HRA Setup to open it.

HRA Setup options differ depending on the type of instrument:

- | | |
|---|-------------------|
| • | HSD Instrument |
| • | Serial Instrument |

HSD Instrument

For an HSD digital pin time domain, the panel looks like this:

HRAM Setup

☒ Manually configure HRAM

Time domain / Protocol port:
DefaultDomain

▲ - Capture Setup

Capture: All

☒ Mask fails until specified event

Halt when: Opcode

Trigger on: FirstCycle

☒ Trigger after specified event

Pretrigger size: 0

HRAM size: 0

▲ - Patgen Event

☐ Cycle count: 0

☐ Module cycle: 0

☐ Time: 0

☐ Loop/repeat: 0

☒ Vector Offset: 10

In module: dual_2x_mod

☐ Label:

Read HW

HRAM Setup Panel, Digital Pin Time Domain

To enable the setup panel, check the box Manually configure HRAM. Otherwise, the panel is not enabled and the current HRAM settings are used and displayed.

If there are multiple time domains or protocol ports or a combination of the two, you configure HRAM for each one separately. Select one from the drop-down under Time domain/Protocol port.

To read the current hardware settings, click Read HW. You can use this button to reset to the original hardware settings after you modify the settings.

HRAM settings are described in the following sections.

These properties of the Capture Setup panel specify what is to be captured:

Capture	Automatically set to All. Waveform Display always captures all cycles to keep its time axis consistent.
Mask fails until specified event	Whether to mask any failing vectors until the specified event occurs. Check this box if you want to mask fails until a specified event as defined in the Patgen Event group of parameters.

Halt when

When HRAM capture should halt. The options are:

- **Opcode:** Halt capture on the halt opcode in the pattern or on the implied halt at the end of a set of pattern modules in a single burst.
- **Fail:** Halt on first fail. Note that once the first fail is encountered, the PatGen does not halt for the depth of the pipeline.
- **HRAM Full:** Halt when HRAM reaches its size limit. The default is the size of the HRAM (512 or 1024 DUT cycles depending on your selected mode, single or dual.)
- **CMEM Full:** Halt when capture memory is full.

Trigger on

The cycle on which HRAM capture is to start. The options are:

- **FirstCycle:** Trigger on the first cycle of the pattern.
- **FirstFail:** Trigger on the first cycle that fails.
- **Never:** Do not trigger.

If Trigger after specified event is checked, this option means to trigger on the first cycle or first fail cycle on or after the cycle on which all of the specified events occur.

Trigger after specified event

If checked, wait for the specified events before triggering capture. Specify events using the Pretrigger size and HRAM size properties.

Pretrigger size

The number of cycles to capture just before the trigger cycle. This is useful for capturing enough cycles to see the context of a failure.

HRAM size

The size of HRAM memory in number of cycles. This parameter makes the size of HRAM memory programmable in software:

- Setting a size smaller than the hardware size makes data readback faster. You can try to reduce the size to the minimum needed to capture the data you want.

- A size of zero (0) means to use the maximum number that will fit in hardware.

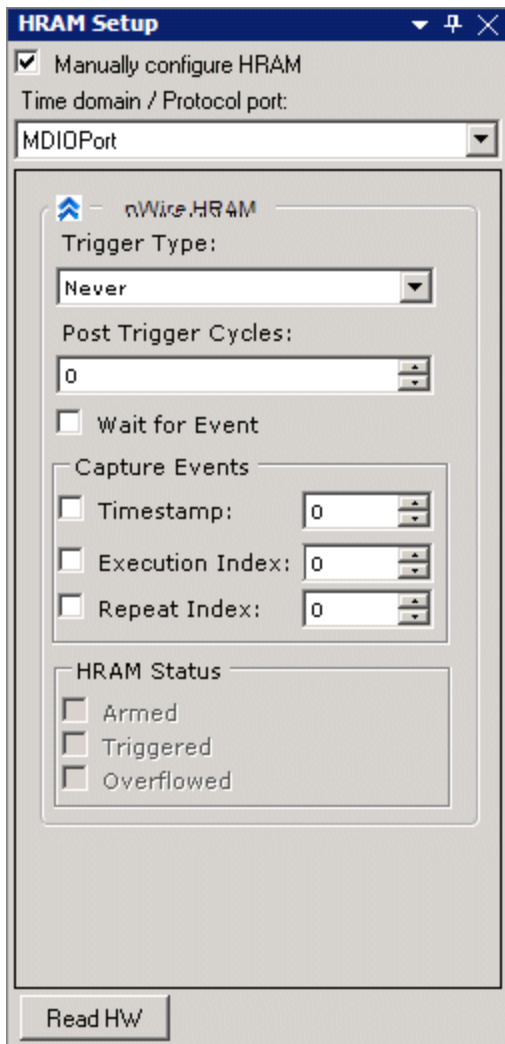
Patgen events items specify the Patgen-related events that can trigger capture. This assumes that the Trigger after specified event box under Setup is checked.

To enable a Patgen event, check its box and then enter the value for the event. You can enable any number of events. All enabled events must occur on the same vector to satisfy the patgen event condition.

Patgen events are as follows:

Cycle count	Count all cycles, up to the specified number.
Module cycle	Count only cycles within a pattern module. The trigger event occurs when any module hits the specified count.
Time	Set the number of time counts that must occur before the PatGen event. The time count is the same as the number of executed master oscillator (MOSC) cycles.
Loop/repeat:	Number of the vector on which to trigger. Lets you define an event as the execution of a particular vector in a pattern module.
Vector Offset:	Number of cycles to capture just before the trigger cycle. This is useful for capturing enough cycles to see the context of a failure.
In module	Select the module. Enabled only if Vector Offset is checked.
Label	Select the pattern label that triggers the event from the drop-down list.

PA instruments also have their own Capture Setup properties. For example, the following is the panel for nWire ports:

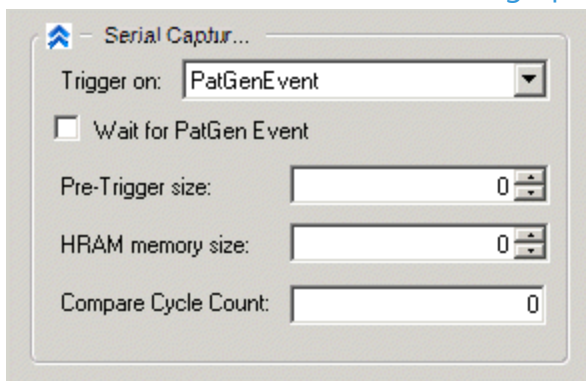


Capture Setup, nWire HRAM Port

For more information, see the section on the PA instruments.

Serial Instrument

For a Serial instrument, the following options are part of the HRAM Setup panel:



Serial Options in the HRAM Setup Panel

Serial Capture options are as follows:

Trigger On: The cycle on which HRAM capture is to start. The options are:

- **Fail:** Trigger on the first compare cycle that fails.
- **PatGen Event:** Trigger on the first compare cycle after the specified PatGen event occurs.

Wait For PatGen
Event

Whether to wait for a PatGen event before trigger occurs.

If the box is not checked (False), capture occurs on the first cycle executed.

If the box is checked (True), capture is delayed until after the event specified under Patgen Events.

Pre-Trigger size:

The number of cycles to capture just before the trigger cycle. This is useful for capturing enough cycles to see the context of a failure.

HRAM memory size:

The size of HRAM memory, in number of cycles. This parameter makes the size of HRAM memory programmable in software:

- Setting a size smaller than the hardware size makes data readback faster. You can try to reduce the size to the minimum needed to capture the data you want.
- A size of zero (0) means to use the maximum number that will fit in hardware.

Compare Cycle
Count:

This parameter is used with two other parameters:

- If Trigger On is set to PatGen Event, delay the trigger until the specified number of compare cycles occur.
- If Wait For PatGen Event is set to True (box is checked), delay the beginning of HRAM capture until the specified number of compare cycles occur.



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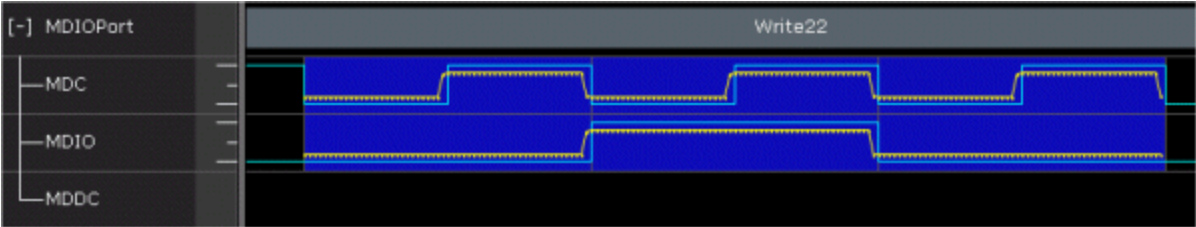
Bursting, Actual Mode, and HRAM : Using Actual Mode

Using Actual Mode

Waveform Display shows ideal digital or PA waveforms based on pattern or module data and timing and levels information. You may want to compare these ideal waveforms to the actual waveforms being sent and received by the digital and PA channels. Actual Mode allows you to overlay the ideal and measured waveforms to perform this comparison.

Actual waveforms are generated by repeatedly rebursting a pattern or PA module while altering the timing and levels for the selected pins, and then examining HRAM to get results. An actual plot takes time, and as results are collected from the hardware, the initial points are plotted. You can stop the collection of Actual Mode data as soon as a waveform is sufficiently complete. Once execution completes, the original tester hardware settings are restored.

The following shows an Actual Mode waveform on PA pins:



Actual Mode Waveform for PA Pins

The following topics are available:

- Steps for Using Actual Mode
- ActualMode Setup
- Actual Mode on Differential Pins

	Notes on Using Actual Mode
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[Bursting, Actual Mode, and HRAM](#) : [Using Actual Mode](#) : Steps for Using Actual Mode

Steps for Using Actual Mode

[Commands for Actual Mode](#) are on the ActualMode menu. Toolbar buttons are available for some Actual Mode commands.

To use Actual Mode:

1. [Use the ActualMode Setup panel](#) to specify the reburst instrument:

- [Last digital burst](#) for a digital pattern.

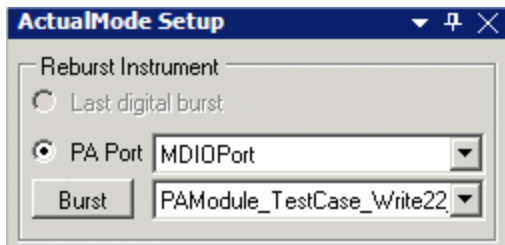
- [PA Port](#). If you select a PA port, there are some additional steps, as described in the next step.

You can also specify nondefault display options. See [ActualMode Setup](#), which describes the [Resolution](#) and [Display](#) controls.

2. [If you selected](#) Last digital burst, go to the next step.

[If you selected](#) PA Port, you must now select the port and module and then burst the module. Waveform Display can access the results of executing the module, but it also needs to know where the transaction came from. This means that you must specify the port and module and burst it.

[When you select](#) PA Port, the following associated controls are enabled:



Actual Mode Setup

- **Port names:** The drop-down next to the PA Port label contains the list of ports that are found in the Burst Explorer. Select the port that you want for the Actual Mode display. If the selected port is not already displayed in the display area, the port is checked in the Burst Explorer and all its function pins are displayed. If any of its function pins is already displayed, no change occurs.
- **Module names:** When you select a port, the drop-down to the right of the Burst button becomes populated with the list of modules that are already loaded on that port. Select a module. Actual Mode can be run only on a single port by bursting a single module at any one time.
- **Burst button:** When you select the port and the module, click this button to burst the module.

You can now go to the next step.

3. Select the range of cells on which you want to superimpose the Actual Mode plot. You can select pins or ports. You cannot select pin groups or pin lists. You can select contiguous cells (by selecting one cell and dragging across the cells you want) or noncontiguous cells (by using the Ctrl key while selecting individual cells). You can also select an entire lane by clicking the row header.

If there is already an Actual Mode waveform on the selected cells, it is cleared before the new Actual Mode waveform is drawn.

For PA, Actual Mode can be run only on cells that represent an executed transaction. Time between transactions is represented as an "Idle" frame. You cannot run Actual Mode on cells in an Idle frame. If you include them in an otherwise valid range, nothing appears in those cells.

4. You can start Actual Mode in any of the following ways:

-  **Run Actual Mode** Use the toolbar button.
- Use the ActualMode>Run Actual Mode command.
- Use the Run button on the ActualMode Setup panel.
- Right-click in the selected cells in the display area and select Run Actual Mode from the context menu.
- Right-click the row header. This selects the entire lane. Select Run Actual Mode from the context menu to run Actual Mode on the entire lane.

The button and commands are enabled only when you have selected a valid range of cells. In addition, for PA, they are enabled only when you have selected and burst the PA module and then have chosen the range of cells.

5. Waveform Display keeps rebursting the pattern or PA module, displaying the samples as they are obtained. If you let execution run, it will terminate when the waveform has the desired resolution and is finished.
6.  **Halt Actual Mode** If you do not need to see the complete waveform, use the Halt command or button to stop the rebursting.
7. Once Actual Mode is finished, you can clear the drawn actual waveforms from the idealized waveform. You can select specific cells to clear and use Clear, or you can use Clear All to clear all actual waveforms in the display. The commands are available on the ActualMode menu and on the context menus.

Refreshing Waveform Display or rebursting a pattern clears all Actual Mode waveforms.

8. You can use the ActualMode>Export Last Actual Mode Run command to save the most recent Actual Mode run as a .wav file in the program directory. The filename looks like this:

AM_DspWave_<pin_name>_StartTime_<time>_ns

If multiple pins are selected, there is one .wav file per pin. The time is the point on the horizontal time axis where the selected cells begin.

You can open the .wav file in WaveScope.

9.

If cursors are enabled, you can have the horizontal cursors indicate the maximum and minimum voltages:
- a.

Right-click either horizontal cursor and right-click again.
- b.

From the context menu, select Snap to Actual Mode Min/Max Voltages.

See Using Cursors.

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WaveformDisplayBurstHRAM.3.11

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Bursting, Actual Mode, and HRAM : [Using Actual Mode](#) : ActualMode Setup

ActualMode Setup

The ActualMode Setup window lets you specify some required information plus some nondefault display options. If the window is not visible, select [View>Actual Mode Setup](#).

ActualMode Setup

Reburst Instrument

☒ Last digital burst

☐ PA Port

Burst

Resolution

Timing:

AutoScaleLo

625.00 ps

Voltage:

AutoScaleLo

222.22 mV

Display

Trace color

Secondary trace color

☒ Draw connecting line

Halt

Run

ActualMode Setup Window

The ActualMode Setup panel has the following sections:

- Reburst Instrument
- Resolution

- Display

Reburst Instrument

To run Actual Mode, the selected waveform must be reburst over and over again while the levels and timing are changed to gather the Actual Mode data points. You must specify what to reburst to generate the waveform using one of the buttons under **Reburst Instrument**:

Last digital burst	Actual Mode uses the last digital pattern that was burst. Default.
PA Port	If the waveform selected is on a Protocol Aware function pin, and if the transactions to execute come from a Protocol Aware module, then select this button. When this button is selected, three associated controls are enabled. See Steps for Using Actual Mode.

If the Burst Explorer contains digital pins, then Last digital burst is selected by default. Otherwise, if the Burst Explorer contains only the Other node and contains only PA ports, then PA Port is selected by default.

Resolution

The Resolution section of the ActualMode Setup panel lets you specify the timing and voltage resolution to use when running Actual Mode.

The Timing resolution affects the horizontal axis and determines how far apart in time two consecutive samples are taken. The options are:

AutoScale	Computes a timing resolution that spans two pixels.
AutoScaleLo	Computes a timing resolution that spans four pixels. Default.

Time samples are not taken in strict increments. Instead, Actual Mode samples are first taken at larger multiples of the specified resolution. These results are drawn at regular intervals drawing the waveform, coarsely at first, and more finely as finer resolution points are filled in. This permits an early view of the complete waveform. You can halt Actual Mode before it completes if you do not need to see the entire waveform filled in.

Voltage resolution affects the vertical axis and determines the number of search bins used when trying to determine where a sample point lies at a specific point in time. The options are:

AutoScale	Computes a voltage resolution that spans two pixels.
-----------	--

AutoScaleLo	Computes a voltage resolution that spans four pixels. This lower resolution allows the Actual Mode waveform to be drawn more quickly. Default.
EdgeMode	Computes a voltage resolution that only draws sample points on High, Low, and mid point voltage levels. Running Actual Mode using Edge Mode instead of Levels Mode allows you to visualize the general wave shape quickly.

Text boxes below the timing and voltage resolution mode combo-boxes display the actual timing and voltage resolutions:

Resolution Fields

Resolution values are displayed change every time you change the timing or voltage resolution mode, or whenever you change the zoom factor over the selection, and possibly also when the selection changes (if the period of the first cycle of the selection is not the same as the previous selection).

Resolution text boxes are also editable. When you specify a specific resolution, the zooming factor is automatically adjusted to maintain the relationship between the resolution mode and the resolution value.

Display

Display controls are found at the bottom of the ActualMode Setup panel:

Trace color	You can set the color for an Actual Mode trace. The default color is yellow. To specify another color, click this button to invoke the standard Windows color chooser and select the desired color from the pop-up control. The currently selected color is displayed next to the button.
Secondary trace color	Actual Mode on a differential pin draws two waveforms. This is the color to be used for the second waveform. See Actual Mode on Differential Pins . Note that the secondary color can be set to Transparent to hide the complementary signal.
Draw connecting line	If this check box is cleared, Actual Mode draws only the sample points on top of the waveform. If this box is checked, Actual Mode draws a line

to connect all sample points. By default, this box is checked.

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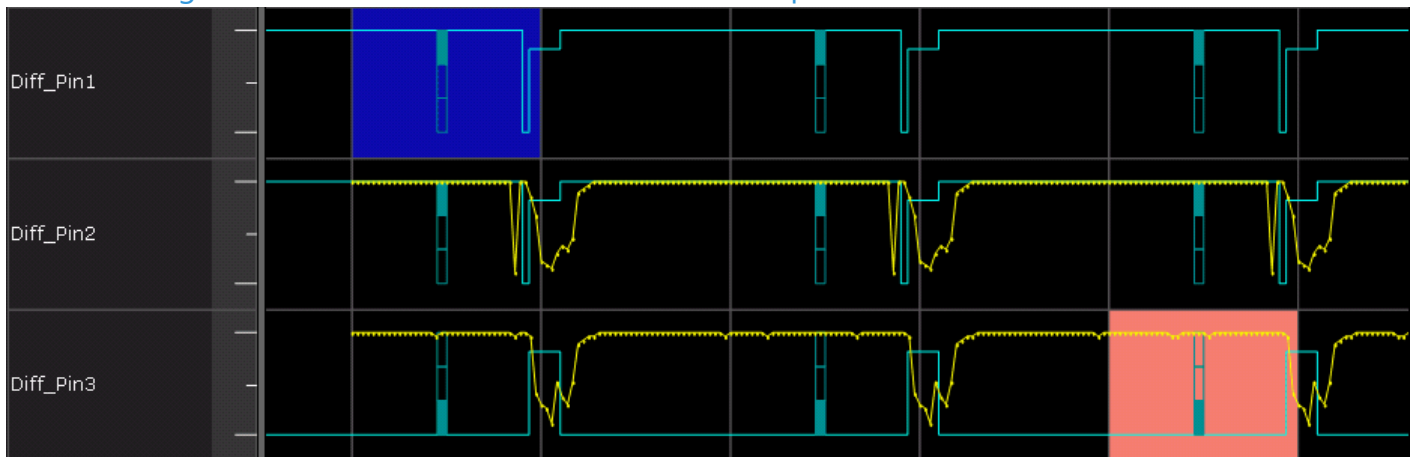
WaveformDisplayBurstHRAM.3.12



Bursting, Actual Mode, and HRAM : [Using Actual Mode](#) : Actual Mode on Differential Pins

Actual Mode on Differential Pins

The following shows Actual Mode as run on differential pins:



Actual Mode Showing Differential Pins

The display uses the main Trace color and the Secondary trace color, as specified on the ActualMode Setup panel. See [ActualMode Setup](#).

A true differential pin supports `<pin>.a` (true) and `<pin>.b` (complementary). Waveform Display shows each half as a separate lane, with the `.a` and `.b` extensions. You cannot run Actual Mode on only one half of a differential pin. If you select cells on either `.a` or `.b`, the other half is automatically selected as well.

Differential pins are handled by drawing the waveform on both halves. The calculated value is displayed, instead of showing single-ended measurements. Differential pins have a lower effective amplitude range than single-ended pins, as determined by the differential comparator.



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WaveformDisplayBurstHRAM.3.13



Bursting, Actual Mode, and HRAM : [Using Actual Mode](#) : Notes on Using Actual Mode

Notes on Using Actual Mode

This topic describes specific situations regarding Actual Mode, as well as apparent anomalies that may arise when using Actual Mode.

Pattern Repeatability

For Actual Mode to work, a pattern must be 100% repeatable. To generate the Actual Mode display, Waveform Display repeatedly executes the pattern while altering the timing and levels for the selected pins. If the execution is not exactly the same for all these repetitions, Actual Mode fails. If a match loop results in a different number of loop executions on various repetitions, the cycle counts will differ. As a result, the pattern is not 100% repeatable and Actual Mode fails.

If pattern flow depends on a decision based on pass/fail activity for a specific pin, Actual Mode will not work if that pin is included in the Actual Mode display. Because timing and levels for the pin will be altered, the pin will pass on some repetitions and fail on others, and the pattern will not be 100% repeatable.

Excluding Code From Actual Mode

You may have code that you do not want to execute while running Actual Mode. For example, you may have a post-pat interpose function that takes a long time to run and is not needed when using Actual Mode.

Use the following VBT statement to check whether Actual Mode is running and then to conditionalize execution:

If Not TheExec.Flow.IsRunningMode(ActualMode) Then

See [IsRunningMode](#) in the Flow section of the VBT Syntax Reference.

KeepAlive Mode

For most digital instruments, Actual Mode does not work with a pattern running in keepalive mode. Check the documentation for the specific digital instrument to see if there are steps to make Actual Mode work with keepalive.

Most digital instruments do not support changing levels and timing in keepalive mode. For this reason, Actual Mode changes levels and timing while rebursting a pattern and therefore cannot work with keepalive.

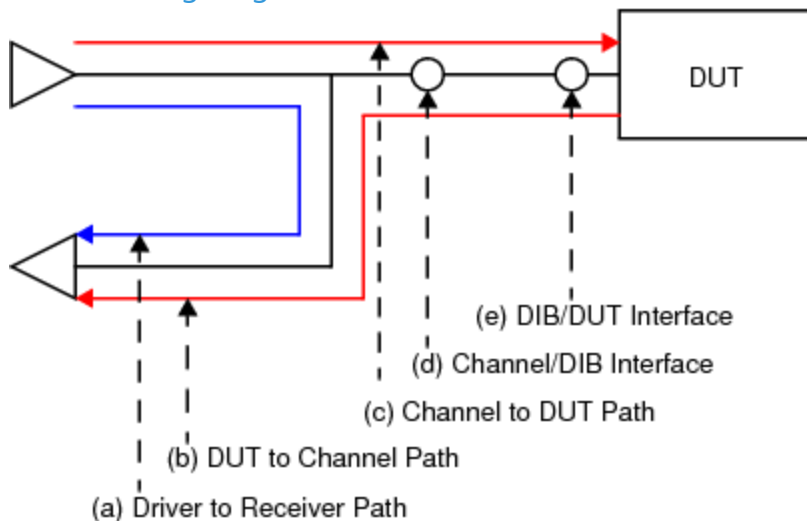
Reflected Signal from Unterminated Circuit or Impedance Mismatch

A circumstance can arise when running Actual Mode on an unterminated drive channel or a circuit for which there is impedance mismatch. In this situation, an additional pulse may appear in Actual Mode that does not appear in the ideal waveform. This is the expected behavior for Actual Mode, in that it is accurately depicting what is happening on the transmission line.

The reason is as follows. If you have an unterminated channel or an impedance mismatch, you will see a reflection in addition to the initial early measurement from the short path between the driver and comparator. This causes two pulses to appear where the ideal waveform shows one pulse. The two pulses may be separate or may be superimposed in a way that causes "staircase" behavior. If you are running at higher speeds with a smaller pulse width, you are more likely to see separate pulses, with a corresponding decrease in voltage because the pulse is split between the drive and receive paths.

Note that this situation arises only with drive cycles.

The following diagram shows the situation in more detail:



Digital Channel Structure

Note these points about the figure:

- The digital channel is calibrated to match received data according to path (b).
- Impedance mismatch at interface points (d) or (e) will cause transmission line reflections. The relationship between the reflections' voltage levels is determined by the transmission line equation, shown below. The timing relationship depends on the difference between path (a) and path (b).

The reflection coefficient (ratio) is:

$$\frac{Z_l - Z_o}{Z_l + Z_o}$$

(3-1)

Z_l is the load impedance (~infinity for HiZ).

Z_o is the line impedance (typically 50ohms).

With HiZ (no termination), the waveforms have equal heights (the ratio is 1).

- Because path (a) is used for Actual Mode, and because path (a) is shorter than the calibrated path (b) plus the transmission path (c), the initial pulse from the transmission line appears earlier than the reflected pulse. Note that the transmission path is along path (c) then back on path (b).

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WaveformDisplayControlling.2.01

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Displaying Waveforms

Displaying Waveforms

This section describes the basic Waveform Display functions and how you can control various aspects of the display:
The following topics are available:

- [Display Basics](#)
- [Navigating](#)
- [Display Options](#)
- [Burst Explorer](#)
- [Properties Window](#)

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WaveformDisplayControlling.2.02

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Displaying Waveforms : Display Basics

Display Basics

This section describes the basic Waveform Display. The following topics are available:

- | | |
|---|--|
| • | Parts of the Display Area |
| • | Waveforms and PatGen Data |
| • | Protocol Aware |
| • | Selecting Digital or PA Mode |
| • | Refreshing and Rebursting |
| • | Taking and Viewing Snapshots |
| • | Selecting the Site |
| • | Viewing Messages |

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WaveformDisplayControlling.2.03

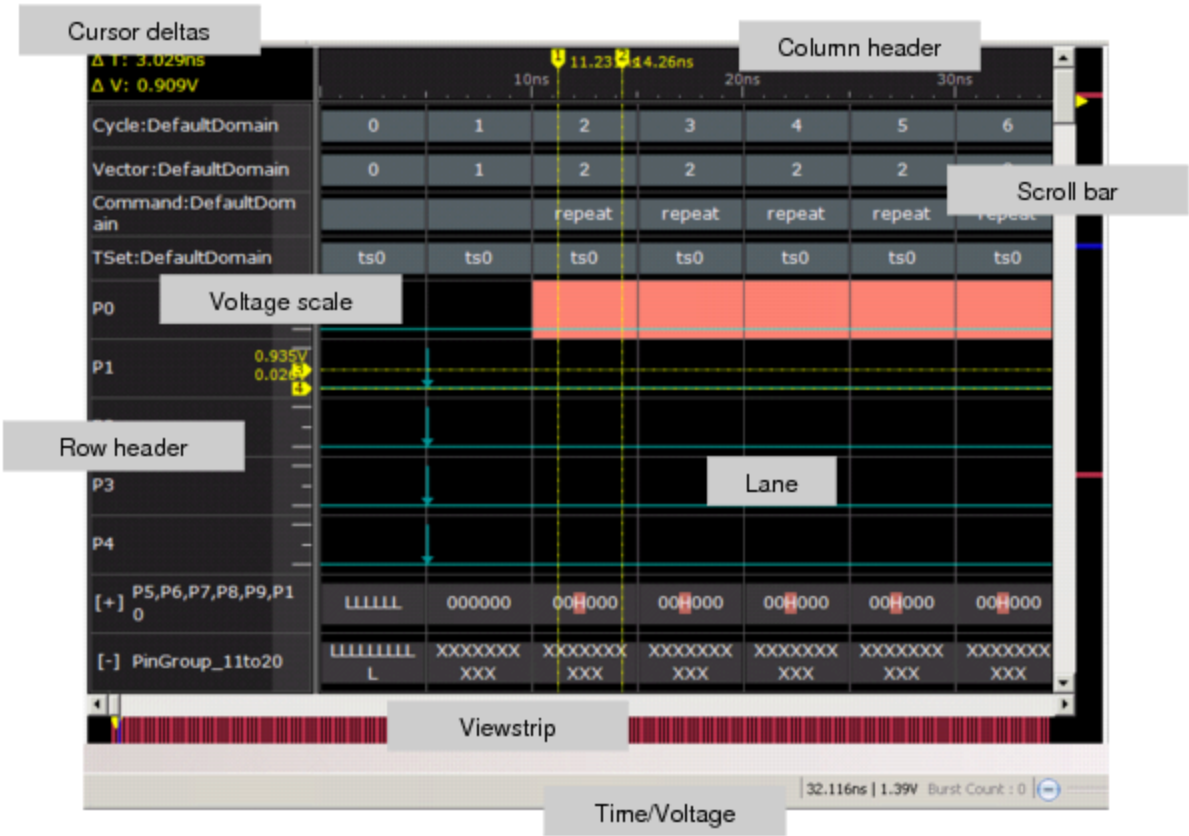
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Displaying Waveforms : [Display Basics](#) : Parts of the Display Area

Parts of the Display Area

The following figure shows the main display area of Waveform Display and labels its most important parts:



Waveform Display, Main Display Area

The display area consists of the following parts:

Lane	A row corresponding to a single pin, group, port, or Patgen element (for example, the timeset or command). A lane spans the entire width of the display area. You can hide or display individual lanes.
Row header	The leftmost part of a lane, giving the name of the element. The context menu lets you hide or display the selected item.
Voltage scale	Horizontal lines indicating voltage levels. This is the rightmost part of the row header. It contains the voltage cursors. Its context menu lets you manage the voltage cursors. See Using Cursors.
Column header	The panel on the top with the time scale and the time cursors. Its context menu lets you manage the time cursors. See Using Cursors.
Cursor deltas	Displays the deltas for the time and voltage cursors. See Using Cursors.
Scrollbars	Scrollbars are visible only when scrolling is possible. See Scrollbars.
Viewstrips	Gives an overview of the entire possible display, indicating the relative position of where failures occurred. Both vertical and horizontal viewstrips are displayed. Double-click in a viewstrip to go to that location. Visible only when scrolling is possible. See Viewstrips.
Time/voltage	The time/voltage coordinates, updated as you move the cross hairs around the display area. See Using Crosshairs.

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WaveformDisplayControlling.2.04

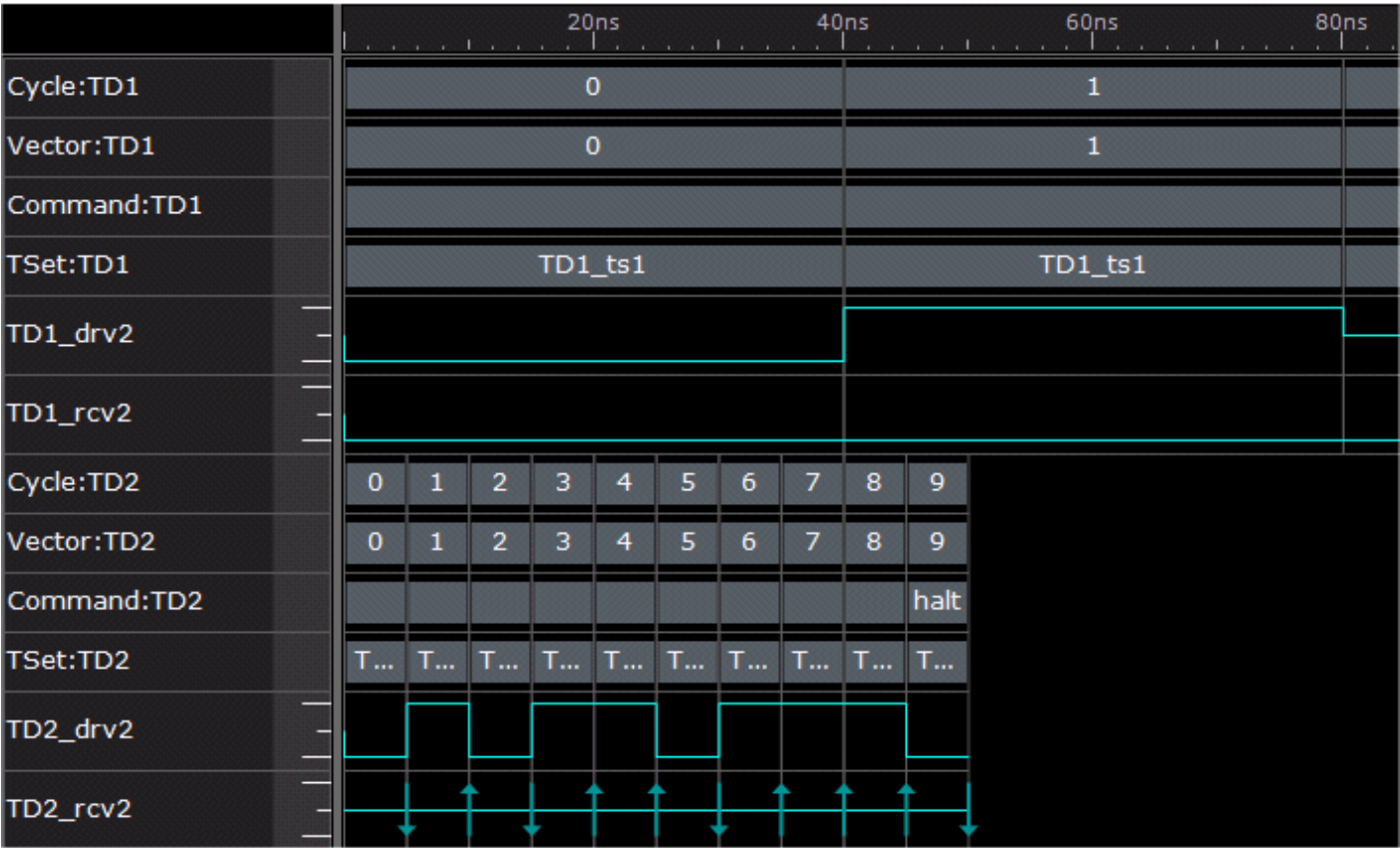
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Displaying Waveforms : [Display Basics](#) : Waveforms and PatGen Data

Waveforms and PatGen Data

A main function of Waveform Display is to represent programmatic waveforms of pins, as shown in the following figure.



Pins in Waveform Display

A waveform resembles an oscilloscope display with each pin's waveform drawn in time order from left to right. The waveform is drawn using HRAM cycle data (programmed data for HSD and actual data for Serial and PA), as well as programmed timing and levels data. Failed cycles on pins have a salmon background color.

You can switch waveforms to symbolic mode and turn the grid and cursors on and off. See [Display Options](#).

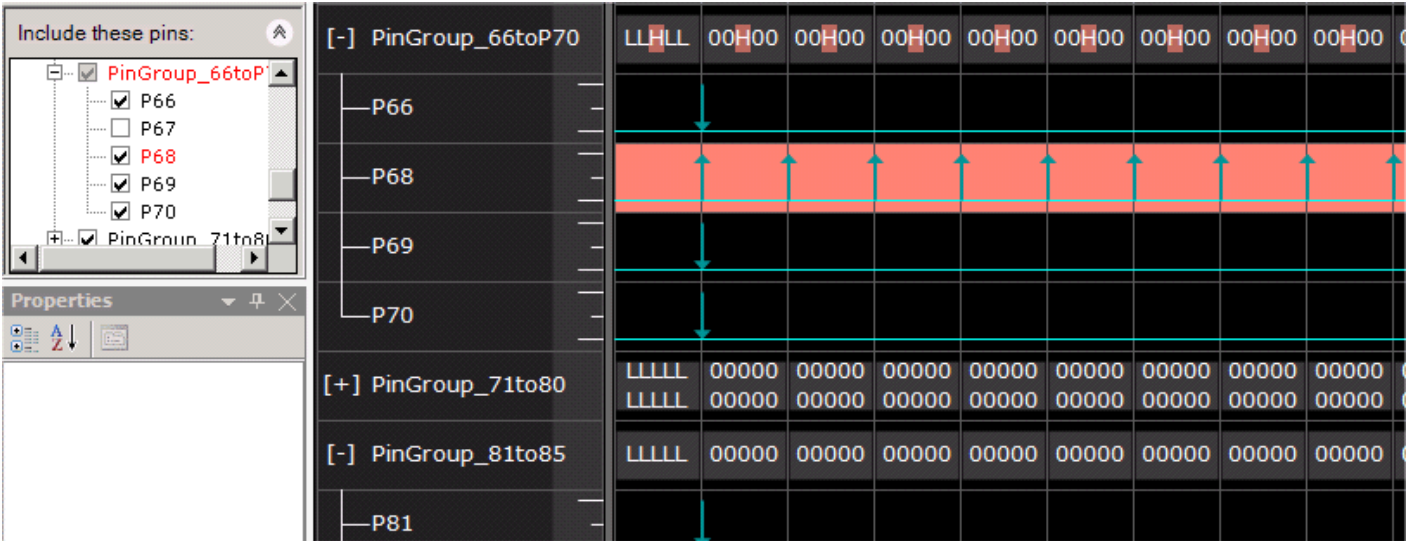
The display also includes Patgen data: cycle and vector number, any command on the vector, and vector tset. In the row header, the time domain name is appended to the Patgen element name so that each row label is unique.

In the row header, the name of the patgen element is appended with the Time Domain name so as to be unique for multiple time domains.

Serial pins can also be displayed in both waveform and symbolic mode, as can serial PRBS and loopback.

Both serial-only bursts and serial pins that are part of an HSD pattern burst can be viewed in Waveform Display. It also allows rebursting of serial patterns using the serial language. Serial Patgen data includes Vector, Cycle, and ModRepeat.

The display area draws a separate lane for each pin group or pin list, as well as a lane for each constituent pin:



Lanes in Waveform Display

The pin group or list lane displays in symbolic mode the data from the individual pins. A failing bit is indicated by the salmon background to the symbolic data character. If one pin in a group or list is unchecked in the Burst Explorer, so that it is not displayed as a lane in the display area, the group or list lane still displays data from the excluded pin. In the above figure, pin P67 is unchecked and therefore not displayed, but the group lane still shows the data for all five pins.

WaveformDisplayControlling.2.05

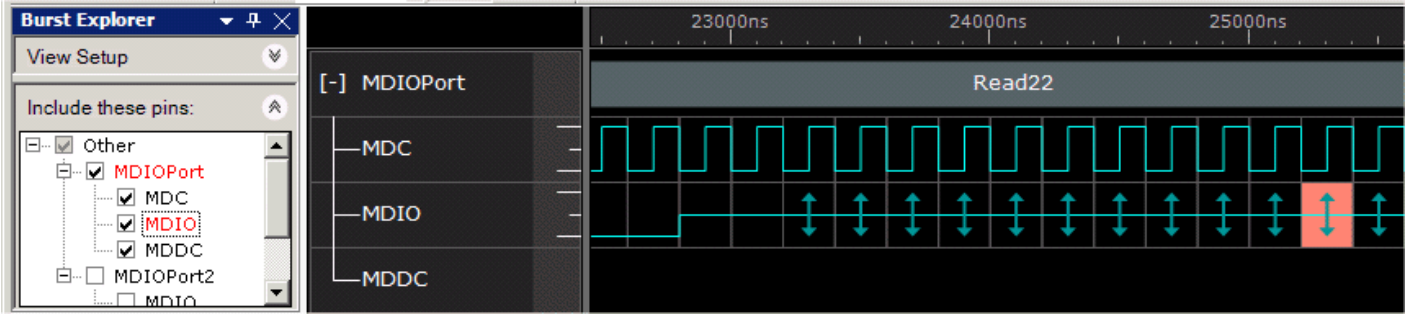
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Displaying Waveforms : Display Basics : Protocol Aware

Protocol Aware

Waveform Display can also show Protocol Aware data:



Protocol Aware Data in Waveform Display

Overall, the display for Protocol Aware has the following structure:

- A PA port is represented in the form of a tree view.
- A PA port title row, comparable to the pin group or list row, contains transaction-level information.
- Pin rows in the PA group contain the individual bits that make up the transaction.

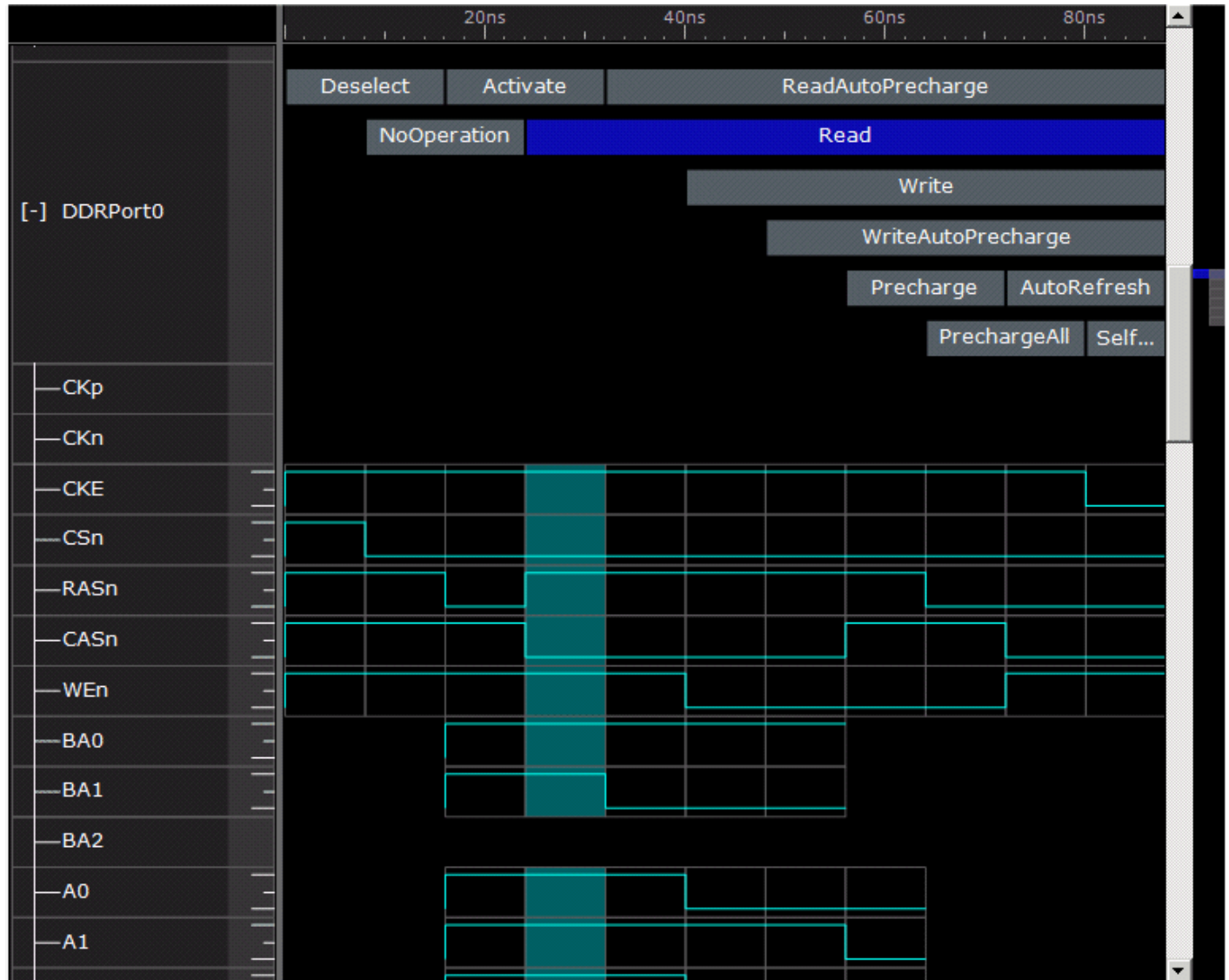
A PA port title row contains the name of the frame. Clicking it highlights all bits that make up fields within the frame.

Function lanes are the rows representing individual function pins that are part of the port. You can switch the PA data to symbolic mode and turn the grid and cursors on and off. See Display Options.

PA ports that are executed from VBT or from a PA module are displayed under the Other node in the Burst Explorer.

To view PA waveforms, you must have Waveform Display open before a test executes the `.IdleWait` or `.HaltWait` command on the ports on which frames were executed. Once you step over `.IdleWait`, Waveform Display receives the data for that frame from IG-XL. If the port on which `.IdleWait` was executed is already on the grid, the tool asks you to manually refresh. The new data replaces existing data for only ports on which `.IdleWait` was called.

The following figure shows DDR waveforms:



DDR Waveforms in Waveform Display

If you click a frame box, such as Read, the display highlights cycles that belong to that frame.

WaveformDisplayControlling.2.06



Displaying Waveforms : Display Basics : Selecting Digital or PA Mode

Selecting Digital or PA Mode

Digital + PA PA Only Digital Only

The toolbar has three buttons that control what data to display. The commands are also available on the Debug menu. The buttons and commands are:

Digital + PA	Displays digital data, both HSD and Serial, and Protocol Aware data. Default.
PA Only	Displays only Protocol Aware data. Does not display digital or serial data, even if a pattern is running.
Digital Only	Displays only digital data, both HSD and Serial.

If you select PA Only, all Patgen and pin items are removed from the Burst Explorer and are therefore not shown in the main display area. Only PA data is shown. This is particularly useful when the digital pattern is looping infinitely in the background providing only a clock for the device. When PA display is enabled, Waveform Display monitors PA transactions. It caches the latest PA transactions executed in VBT (either through test execution or the Immediate window) or in a pattern. For each port, only data provided by the last IdleWait or HaltWait is cached. Monitoring PA increases execution time. If you are interested only in digital or serial data, specify Digital Only to turn off PA monitoring and reduce execution time.



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WaveformDisplayControlling.2.07

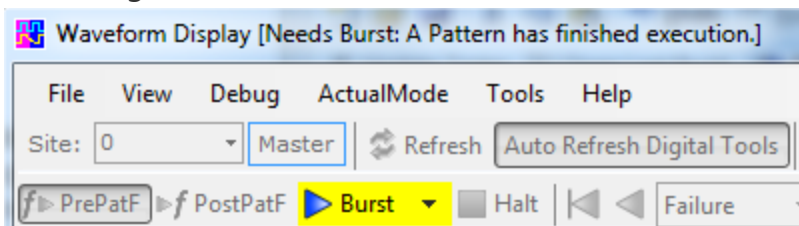


Displaying Waveforms : Display Basics : Refreshing and Rebursting

Refreshing and Rebursting

When you explicitly reburst a pattern in Waveform Display, it automatically refreshes its display. See [Bursting Patterns](#) and [Using Actual Mode](#).

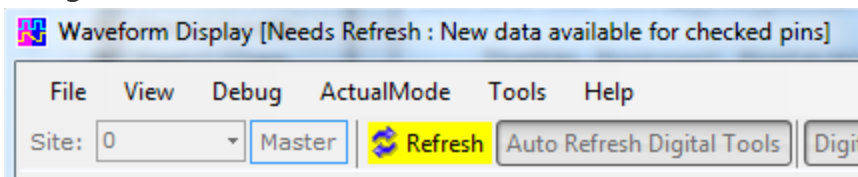
A pattern may be reburst outside Waveform Display: for example, by another tool or by stepping through program code. In this case, the message Needs Reburst is displayed in the window title bar, along with the reason the reburst is needed. The Burst button is also highlighted in yellow:



Needs Reburst Message in the Title Bar

Click [this button](#) or use [Debug>Burst](#) or F12 to display the results of the most recent burst.

When new PA data is received, the message Needs Refresh is displayed in the window title bar, along with the reason the refresh is needed. The Refresh button is also highlighted in yellow:



Needs Refresh Message in the Title Bar

Click [this button](#) or use [View>Refresh](#) or F12 to display the most recent PA data.

By default, Waveform Display automatically rebursts the last burst pattern when you first launch the tool. You can turn this feature off. See [Other Tool Settings](#).

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WaveformDisplayControlling.2.08



Displaying Waveforms : [Display Basics](#) : Taking and Viewing Snapshots

Taking and Viewing Snapshots

Waveform Display lets you take snapshots of the current state of the tool. Later, you can view the snapshots for a visual comparison of tool states.

The [command](#) View>Take Snapshot takes a snapshot of the Waveform Display window. The View>View Snapshot command lets you view the snapshots. It has these options:

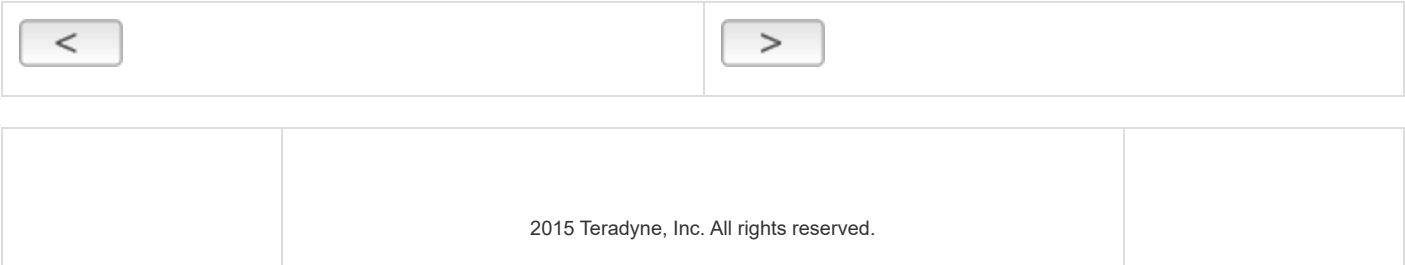
Most Recent	Displays the most recent snapshot.
Other	Brings up a browser to let you select the snapshot to view.

The display uses the default Windows image viewer.

By default, the snapshot is saved to the test program folder. If the default folder is read-only, it is saved to the tmp folder. You are not prompted to specify a folder. The snapshot is saved as a JPEG file with a filename of this form:

Waveform Display_YYMMDDHHMMSS.jpg

This means that the snapshots in the browser will be in chronological order.



WaveformDisplayControlling.2.09

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Displaying Waveforms : [Display Basics](#) : Selecting the Site

Selecting the Site

Site: 0 ▾ Master

Waveform Display shows the data for one site at a time. Use the site selector to specify a site.

If you [disable](#) Master, you can manually select the site to be displayed. If you change the site, Waveform Display autorefreshes to display the data from the new site.

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WaveformDisplayControlling.2.10

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[Displaying Waveforms](#) : [Display Basics](#) : Viewing Messages

Viewing Messages

Messages that do not require an immediate user response are written to the Message Panel. For more information, see [About the Message Panel](#).

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WaveformDisplayControlling.2.11



Displaying Waveforms : Navigating

Navigating

Waveform Display provides the following ways to navigate to specific points in the display area:

- [Goto](#)
- [Viewstrips](#)
- [Scrollbars](#)



WaveformDisplayControlling.2.12



Displaying Waveforms : Navigating : Goto

Goto

 The Goto control on the toolbar lets you go to a specific item in the display. Select what you want to go to using the drop-down list in the combo box. If you specify which item to go to (cycle, time, or vector), the text box is enabled for you to enter a value. The red buttons let you go (from left to right) to the first, the previous, the next, or the last occurrence of the specified item. If you are going to a specific cycle or time, only the Next button is enabled; it takes you to the specified item in the current lane. If the selected item is invalid (for example, if you select “Frame” in a non-protocol display), no buttons are enabled.

The options in the combo box are:

Failure	Go to the first/previous/next/last failing cycle.
Frame	Go to the first/previous/next/last protocol frame.
Cycle	Go to the specified cycle number in the lane of the current selection. Enter the cycle number in the text box and use the Next button.
Time	Go to the specified time. Enter the time in nanoseconds in the text box and use the Next button.
Vector	Go to the first/previous/next/last vector with the specified number. Enter the number in the text box. Because of repeats and loops, multiple vectors can have the same number.

If multiple lanes are part of the selection, the GoTo applies to the lane of the first item selected. If you are at the first or last item, Previous and Next do not cycle back to the end or beginning of the display.

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WaveformDisplayControlling.2.13

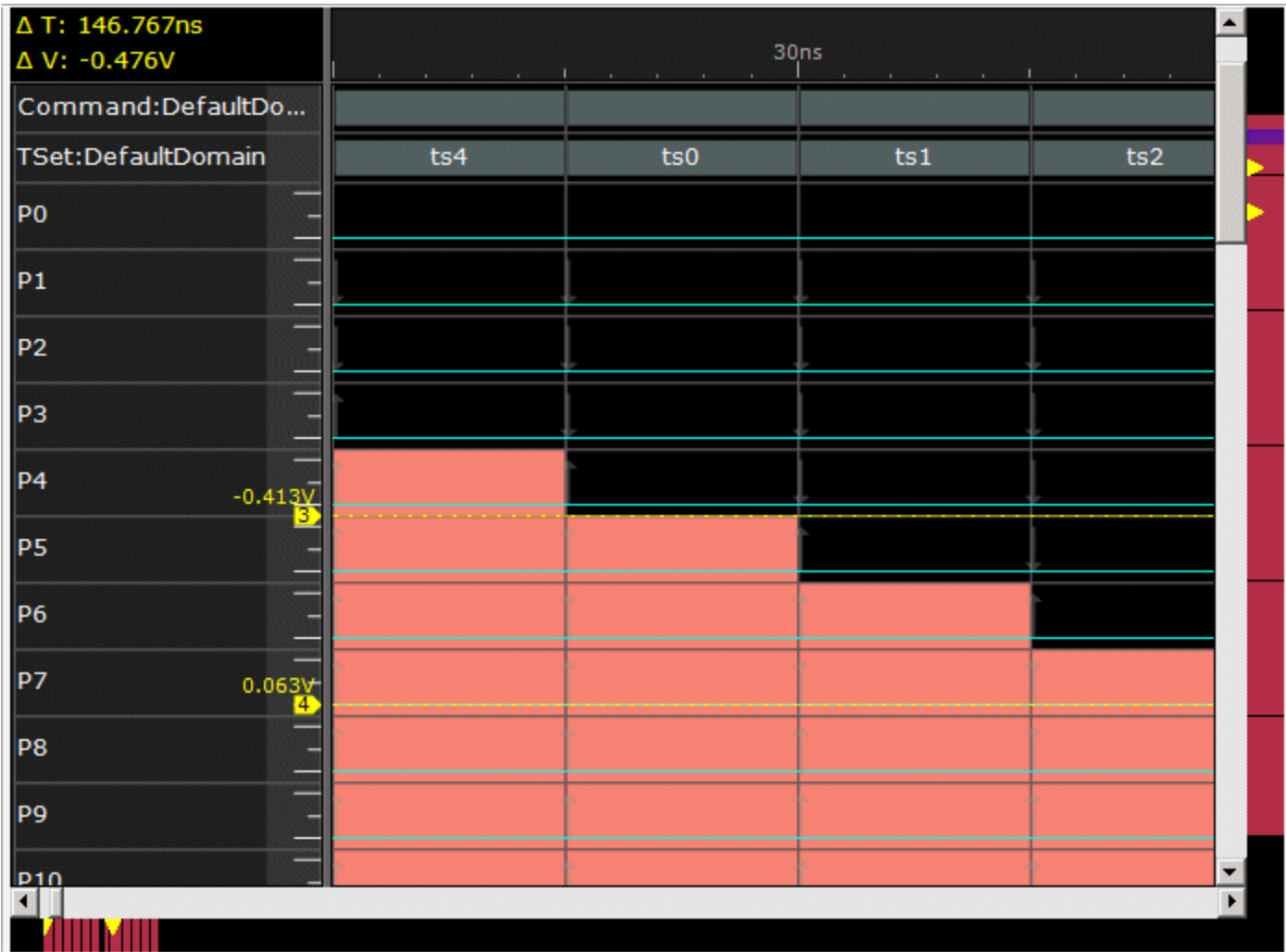
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Displaying Waveforms : Navigating : Viewstrips

Viewstrips

If the data to be displayed extends beyond what is visible in the display area, a horizontal or vertical viewstrip or both become visible:



Vertical and Horizontal Viewstrips in the Display

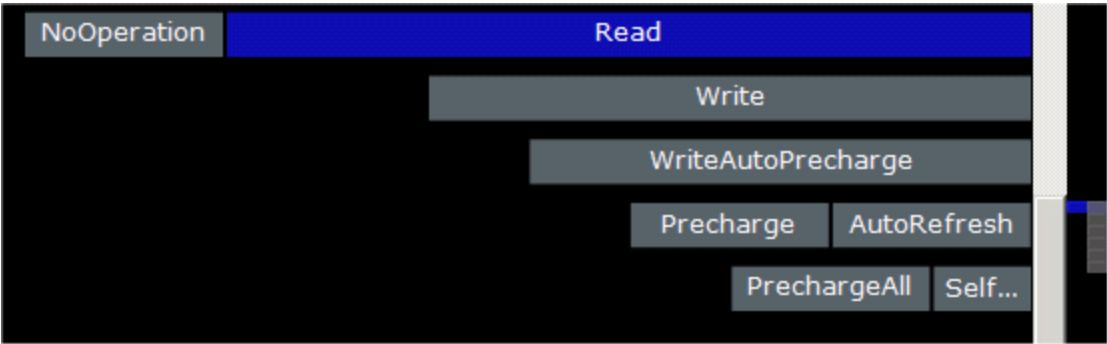
The viewstrips are the colored strings to the right of and below the scrollbars.

A viewstrip gives an overall view of the data, specifically of all captured failures across the horizontal time scale and the vertical pins or ports. They show the relative positions of the failures in relation to the entire span of data.

You can double-click at any point in a viewstrip to navigate to that position in the data. Clicking in the horizontal viewstrip scrolls only on the horizontal time axis. Clicking in the vertical viewstrip scrolls only on the vertical lane axis.

The following colors are used to indicate items in the viewstrip:

- A red band indicates a cycle that failed at that location. If you hover over a failure, a tooltip displays the fail cycle number.
- A gray band represents a frame that occurred at that location. The tooltip displays the frame name.



Frames in Viewstrips

- A blue line indicates the current selected position.
- Yellow triangles indicate the current cursor positions.

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WaveformDisplayControlling.2.14



Displaying Waveforms : Navigating : Scrollbars

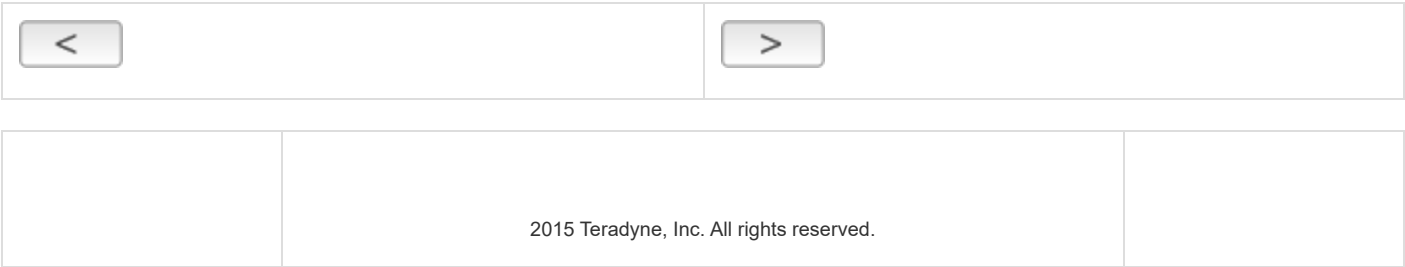
Scrollbars

If the data to be displayed extends beyond the display area, a horizontal or vertical scrollbar or both appear at the right and bottom of the tool.

You can use the arrows at each end of a scrollbar to move smoothly in that direction. You can left-click a scrollbar to jump a small distance in that direction. If you right-click a scrollbar, the context menu gives you more options:

- Scroll here, that is, to the point at which you clicked on the scrollbar.
- Scroll all the way to the top or bottom or left edge or right edge.
- Scroll one page’s worth up or down or to the right or left.

You can freeze one or more of Patgen lanes to keep them at the top of the display during vertical scrolling. See Freezing the View.



WaveformDisplayControlling.2.15



Displaying Waveforms : Display Options

Display Options

Waveform Display offers a set of controls to customize the wave display area. These are on the View menu or the top toolbar:

- [Using the Grid](#)
- [Using Cursors](#)
- [Using Crosshairs](#)
- [Using Symbolic Mode](#)
- [Using Zoom](#)
- [Freezing the View](#)
- [Other Tool Settings](#)



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WaveformDisplayControlling.2.16

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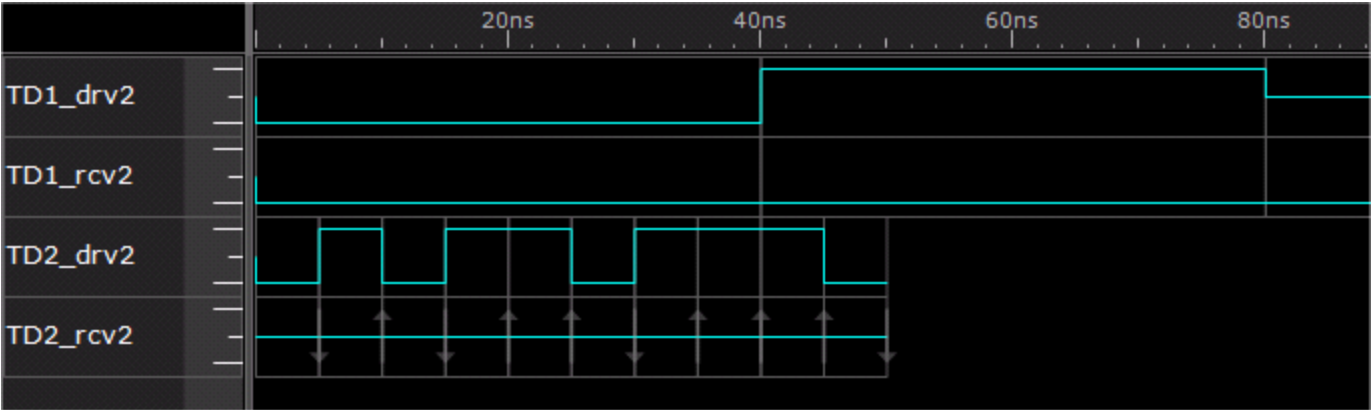
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Displaying Waveforms : [Display Options](#) : Using the Grid

Using the Grid

 [This button or command](#) (View>Grid) turns the grid on and off. The grid's horizontal lines are along pin-row boundaries, and the vertical lines are along cycle boundaries.

[When pins belonging to different time domains are displayed, the vertical lines do not align:](#)



Grid in the Display

Some waveforms may be sparse: for example, when viewing transactions for DDR. In these cases, no grid lines are drawn at points where no data is available.

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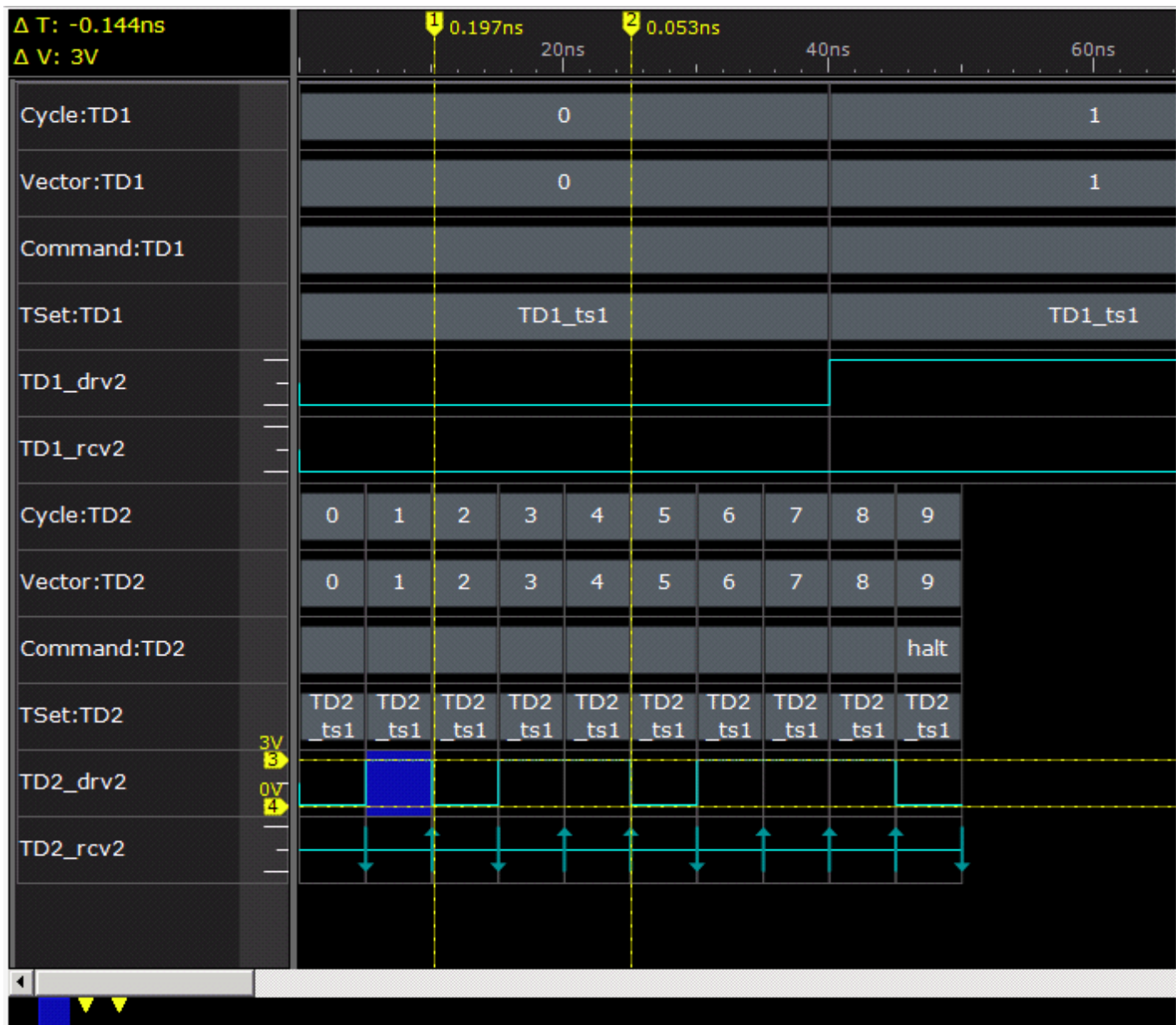
WaveformDisplayControlling.2.17



[Displaying Waveforms](#) : [Display Options](#) : Using Cursors

Using Cursors

 **Toggle Cursors** This button or command ([View>Cursors](#)) turns the cursors on and off:



Horizontal and Vertical Cursors in the Display

The time scale includes two horizontal cursors numbered 1 and 2. The voltage scale includes two vertical cursors numbered 3 and 4. All cursors are either on or off. You cannot enable a subset of the cursors.

The values of the horizontal time cursors are relative to the currently selected lane. For example, in the display above, the time values are relative to the TD2_drv2 lane. If no cells are selected, the cursor corresponds to cells in the first lane. If multiple cells are selected, the cursor values correspond to the last cell selected.

The viewstrips indicate the relative position of the cursors using yellow triangles. One way to go to a cursor that is not visible is to double-click the yellow triangle in the viewstrip.

The upper left cell contains the cursor deltas: the distance between the timing cursors and the difference between the voltage cursors. The voltage delta is the arithmetic difference between the two positions, even if they are in lanes for different pins.

To move a cursor, click the numbered cursor handle and drag it to the position you want.

To see more cursor control options, right-click in the cursor areas. These are the timing column headers at the top of the display area and the voltage scales, which are at the right of each row header and contain the horizontal voltage marks. The options appearing on the context menu depend on whether cursor are turned on and whether you are right-clicking in the timing scale or the voltage scale:

Toggle All Cursors Off/On	Same as the button and command.
Bind Cursors 1 And 2/3 And 4	Cause the bound cursors to move in unison, so that the delta between them remains constant. The two bound cursor handles are connected by a yellow line.
Scroll To Cursor 1/2/3/4	Scroll the display to show the indicated cursor.
Move Cursor 1/2/3/4 Here	Move the indicated cursor to the current position on the time scale (1 or 2) or voltage scale 3 or 4).

If you have run Actual Mode, the context menu includes an additional command: Snap to Actual Mode Min/Max Voltages. Selecting this command moves the horizontal cursors to indicate the maximum and minimum voltages.

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WaveformDisplayControlling.2.18

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Displaying Waveforms : [Display Options](#) : Using Crosshairs

Using Crosshairs

As you move the crosshair across the waveform display area, the time and voltage coordinates are continually displayed and updated in the status bar:



Time and Voltage Coordinates in the Status Bar

The precision is up to four significant figures. The units are ns or ms (nanoseconds or milliseconds, depending on precision) for time and V (volts) for voltage.

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WaveformDisplayControlling.2.19

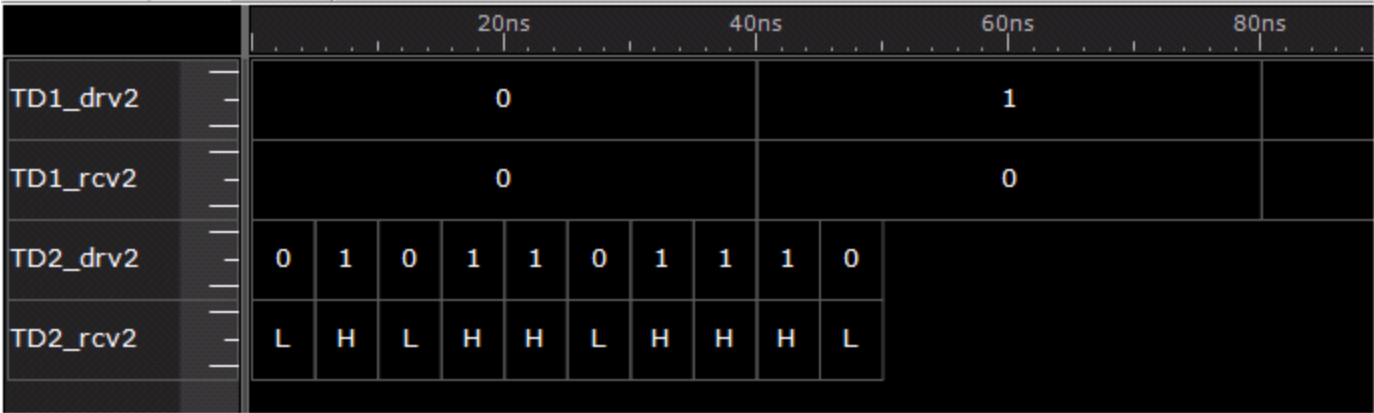
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Displaying Waveforms : [Display Options](#) : Using Symbolic Mode

Using Symbolic Mode

 This button or command ([View>Symbolic](#)) toggles between displaying data in waveform mode and symbolic mode, that is, showing the pin data values. The following shows figure the symbolic representation of the same waveforms shown in [Using the Grid](#) and [Using Cursors](#):



Waveform Shown in Symbolic Mode

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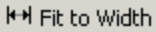
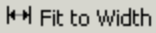
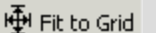
WaveformDisplayControlling.2.20



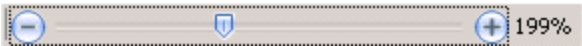
Displaying Waveforms : Display Options : Using Zoom

Using Zoom

The View>Zoom command has options that let you zoom in on the currently selected range of cells. The options are also available as toolbar buttons. Select a range of cells in the display, and then select one of these options:

Fit to Width	 Fit to Width	Expand the selection horizontally so that it fills the width of the display area. The height is unchanged. This option is also available if you select a range of cells, right-click, and select it from the context menu.
Fit to Height	 Fit to Height	Expand the selection vertically so that it fills the height of the display area. The width is unchanged.
Fill To Full Grid	 Fit to Grid	Expand the selection so that it fills both the height and width of the display area.

In addition to the Zoom command and buttons, there is one zoom track bar at the bottom of the window and one at the right side of the display window. The bottom horizontal track bar looks like this:

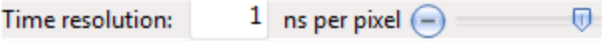


Horizontal Zoom Track Bar

The horizontal track bar lets you increase or decrease the horizontal time scale of the display. Only the horizontal time scale is affected, not the vertical height of the lanes. The vertical track bar lets you increase or decrease the vertical voltage scale.

You can grab and drag the slider or click the track bar to move the slider in that direction. The maximum horizontal zoom is 10000%, and the minimum zoom is 12%. The maximum vertical zoom is 2000% and the minimum zoom is 79%.

 **Zoom to 100%** To return to the default 100% horizontal and vertical zoom, use this button or the View>Zoom to 100% command. This operates on the entire display, not only on any selected cells. The control to the left of the horizontal track bar lets you set the time scale:



Time Resolution Control

Time Resolution specifies how many nanoseconds are to be represented per pixel on the time axis of the display. The default is 1. A higher resolution gives a more compressed display on the horizontal axis. Valid values range from 0.001 to 100. You can also set this using DataTool settings. See [Other Tool Settings](#).

			
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WaveformDisplayControlling.2.21

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Displaying Waveforms : Display Options : Freezing the View

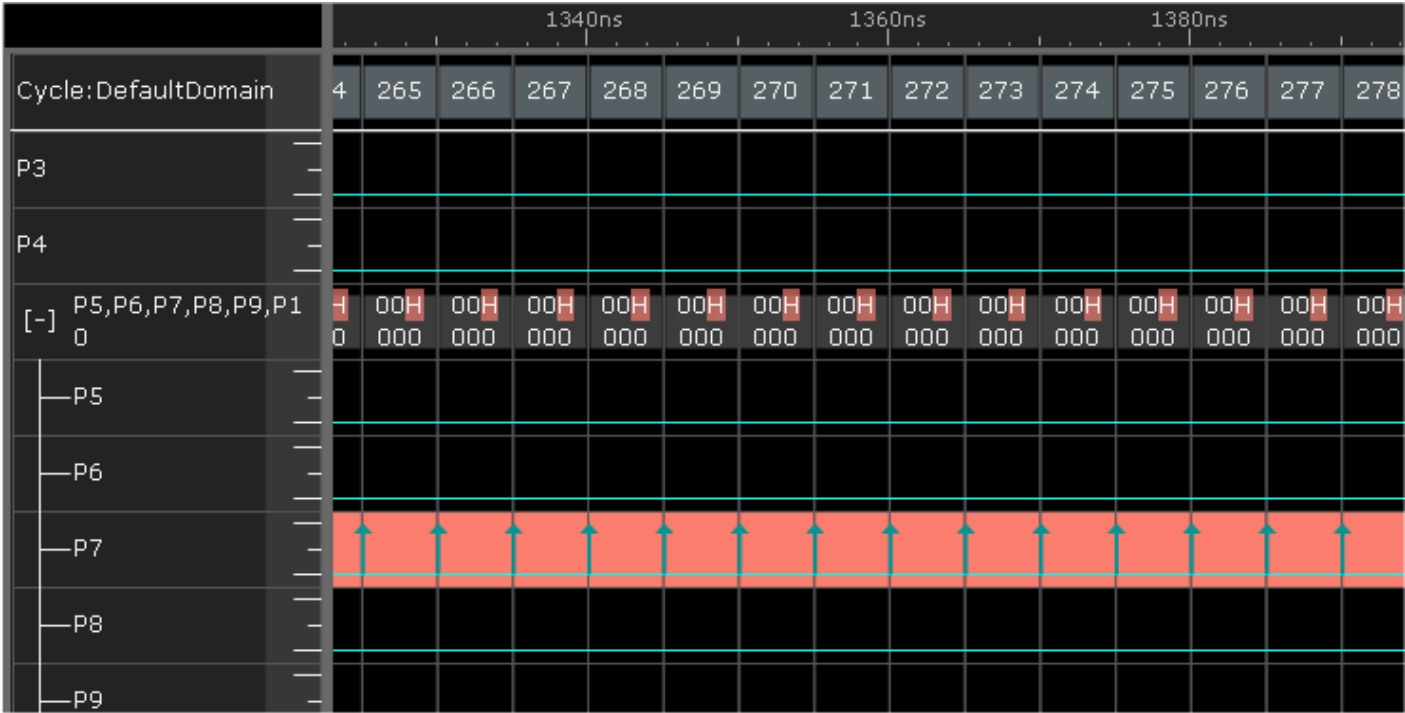
Freezing the View

You can freeze the display so that one or more of the Patgen lanes remain visible during vertical scrolling. The Patgen lanes are Cycle, Vector, Command, and TSet.

To freeze a Patgen lane:

1. Select the Patgen lane header and right-click.
2. Select Freeze from the context menu.

The lane is moved to the top of the display with a white line under it:



Patgen Lane Frozen in the Display

You can add other Patgen lanes to the freeze section by following the same steps. Each new frozen lane is added after the previously frozen lanes. The white line appears after all frozen lanes.

To unfreeze a specific lane:

- | | |
|----|--|
| 1. | Select a lane header and right-click |
| 2. | Select Unfreeze from the context menu. |

To unfreeze all lanes:

- | | |
|----|---|
| 1. | Select the header of any frozen lane and right-click. |
| 2. | Select Unfreeze All from the context menu. |

		
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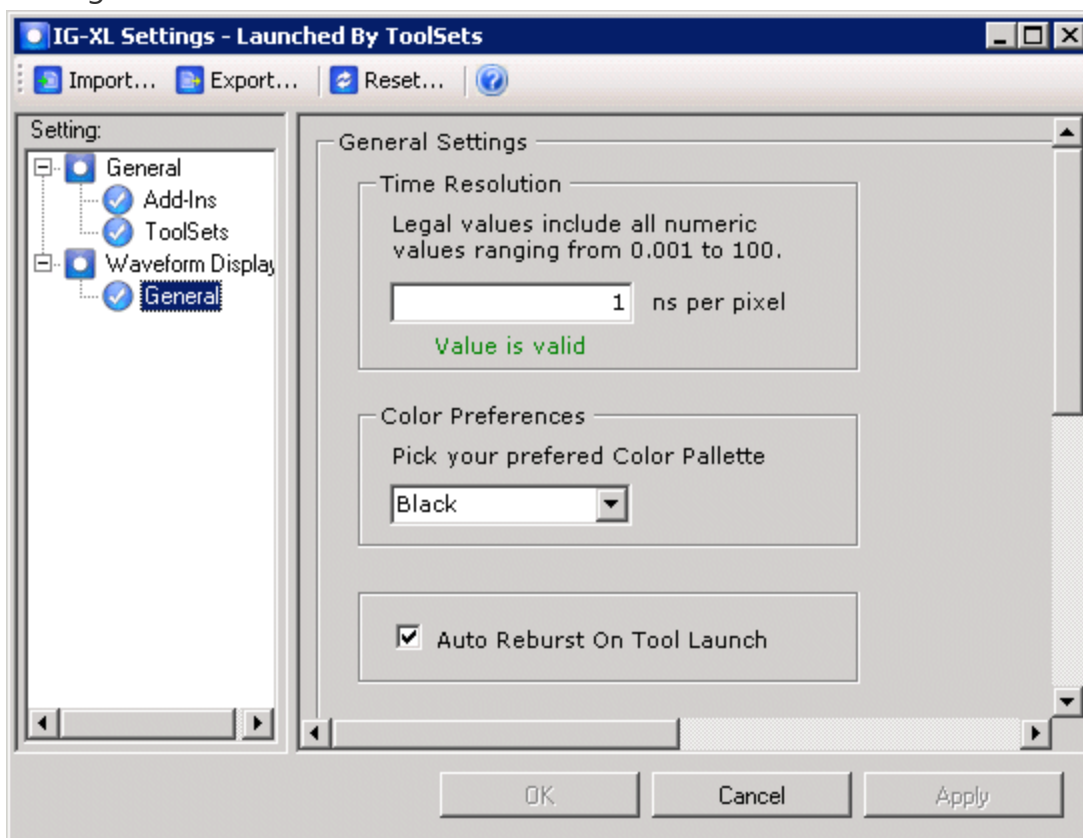
WaveformDisplayControlling.2.22



[Displaying Waveforms](#) : [Display Options](#) : Other Tool Settings

Other Tool Settings

You can specify other display settings. Use the command [Tools>Settings](#). This brings up the IG-XL Settings window:



IG-XL Settings Window

The following settings are available:

Timing Resolution	Specifies how many nanoseconds are to be represented per pixel on the time axis of the display. The default is 1. A higher resolution gives a
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	more compressed display on the horizontal axis. You can also set this on the Waveform Display window itself, using the control at the lower right corner. See Using Zoom.
Color Preference	Specifies the background color for the display area. The default is Black. You can also select Green, which gives a light green background color.
Auto Reburst On Tool Launch	Automatically refreshes the tool on its launch by rebursting the last pattern.

For more information, see [Using Tool Sets](#) in the DataTool section.

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WaveformDisplayControlling.2.23

	
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Displaying Waveforms : Burst Explorer

Burst Explorer

By default, the Burst Explorer and the Properties window are on the left side of the tool. If they are not visible, you can use commands on the View menu to make them visible.

The following topics are covered:

- [Burst Explorer Treeview Structure](#)
- [Selecting Pins and Ports](#)
- [Using View Setups](#)

	
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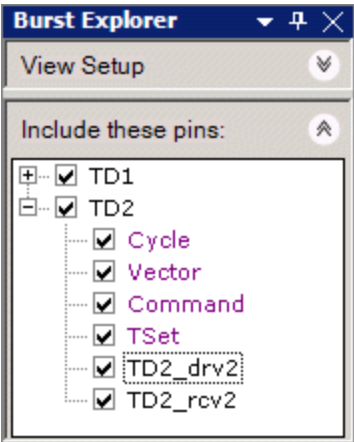
>

Displaying Waveforms : Burst Explorer : Burst Explorer Treeview Structure

Burst Explorer Treeview Structure

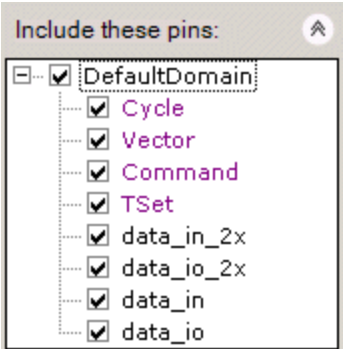
The Burst Explorer uses a treeview structure to display the pins, ports, and other items from the last burst file, as well as their hierarchical relationship to each other.

For pins, the topmost node in the treeview is a time domain. Each time domain is a separate top node:



Burst Explorer, Showing Multiple Time Domains

If multiple time domains are not defined, the Default Time Domain is used:

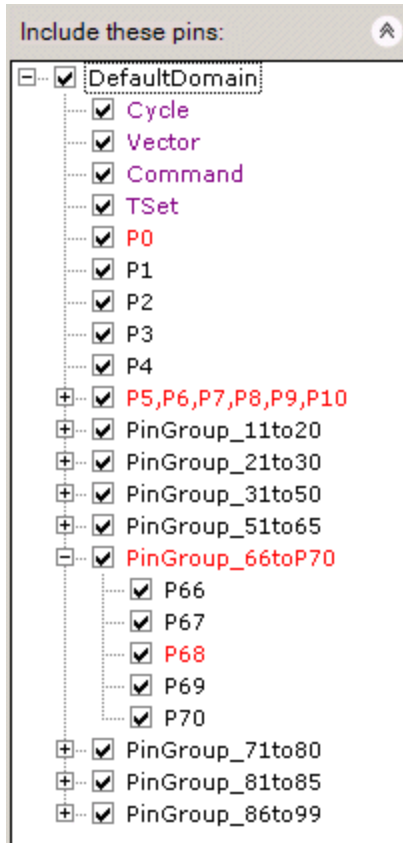


Burst Explorer, Showing Default Time Domain

Under the time domain are non-pin items: cycle, vector, command, and tset. These are shown in purple. Serial patterns have additional non-pin items, also shown in purple.

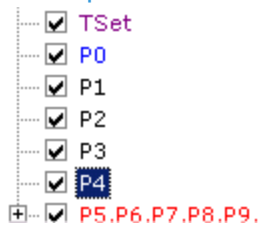
Each pin or pin group is a separate item under a time domain. Each pin group can be expanded.

Failing pins are in red, as are pin groups with failing pins in them:



Burst Explorer, Showing Failing Pins in Red

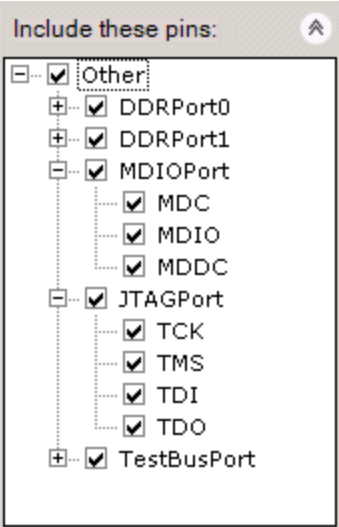
Masked pins are shown in blue:



Burst Explorer, Showing Masked Pins in Blue

Note that you must have checked a pin's checkbox at one point for it to be displayed in red or blue.

For ports, the top-level node is Other, and the ports and their pins are listed beneath it:



Burst Explorer, Showing Ports

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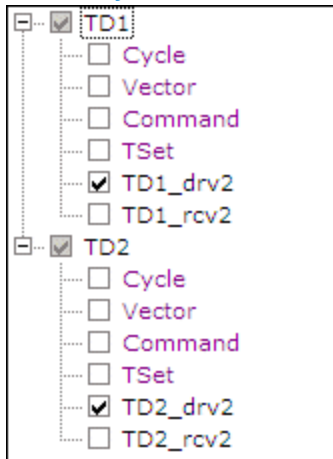
Displaying Waveforms : Burst Explorer : Selecting Pins and Ports

Selecting Pins and Ports

Checkboxes in the Burst Explorer control what items are included in the display. To include an item in the display, check its box. To remove it uncheck the box.

After you refresh a burst, nothing in the Burst Explorer is checked. You must check the boxes you want displayed.

Checking or unchecking a parent node in the treeview also checks or unchecks all its children nodes. For example, checking a time domain box checks all the pin and other boxes that belong to it. If you check a child node, the check box for its parent node is checked but grayed out to indicate that only individual subitems have been checked:



Burst Explorer, Subitems Unchecked

Clicking an item name itself does not select it, although it does show the item's properties in the Properties window. You must check a box to include it in the display.

You can also remove one or all items from the display using a context menu. In the display area, right-click a lane. The menu has these options, among others: Remove and Remove All. If you remove an item in this way, it is also unchecked in the Burst Explorer.

Once you set up the display as you want it, you can save it to view setup. You can later recreate the display by selecting the setup. See [Using View Setups](#).

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WaveformDisplayControlling.2.26

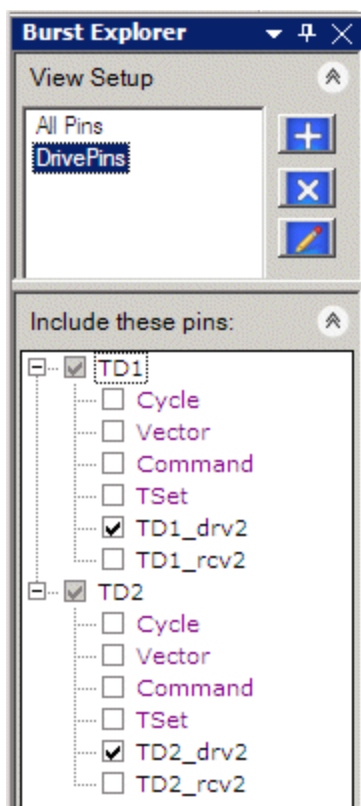


Displaying Waveforms : Burst Explorer : Using View Setups

Using View Setups

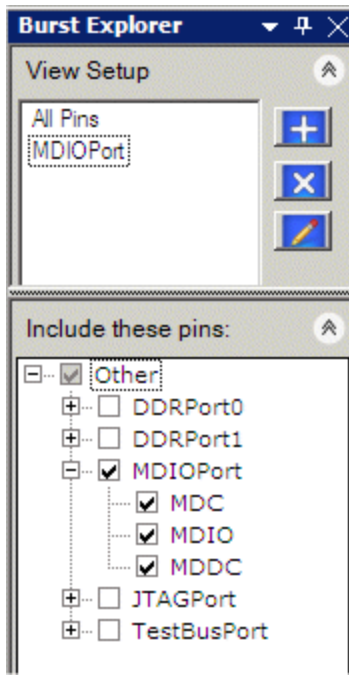
You can use the Burst Explorer to create view setups. A view setup lets you select items that you want included in the rows of the display and save the selection as a named setup. You can later redisplay those rows by clicking the name of the setup.

The first example displays only drive pins:



Burst Explorer Showing Only Drive Pins

The second example displays a subset of the available ports:



Burst Explorer Showing Selected Ports

Available setups are listed in the View Setup panel. If this panel is not visible, use View>Burst Explorer to make it visible.

The default built-in setup is All Pins, which is equivalent to checking all boxes in the Burst Explorer panel. You cannot modify or delete this setup.



To create a view setup, select items that you want included by checking their boxes in the Burst Explorer. Then click the Add button. A new string is added to the View Setup panel with the default name "setupN." The name is in edit mode, so you can change it immediately.

Once you add a setup, you can double-click the name to display the selected rows.



To delete a setup, select it in the View Setup panel and click the Delete button. To delete all user-created setups, right-click in the white area of the View Setup panel (not on a setup name) and select Delete All.



To modify a setup, select it in the View Setup panel. Make changes to the check boxes. Then click the Update button.

To rename a setup, select it in the View Setup panel, right-click, and select Rename. Note that Apply, Delete, and Update are also on the right-click menu.

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Displaying Waveforms : Properties Window

Properties Window

The Properties window displays the properties of the currently selected pin or port. The pin or port can be selected in the Burst Explorer or in the display area. There are differences in the way pins and ports are displayed, as described below:

- Pin Properties
- Protocol Properties

If the Properties window is not visible, use View>Properties to display it.

The Properties window can be divided into categories. The first category is always General. The other categories depend on the kind of item selected.

You can specify how the properties are to be displayed using the buttons at the top of the Properties window:

-  Categorized, for example, General, Levels, Timing
-  Alphabetical, without category dividers



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WaveformDisplayControlling.2.28



Displaying Waveforms : Properties Window : Pin Properties

Pin Properties

The Properties window displays the properties of the selected pin, pin group, or pin list.

You can modify some properties before rebursting to do basic debug. Properties that you can change are in black. Properties that you cannot change are grayed out. For example, you can change a pin's voltage levels, but not its name. You can change a pin's timing edge or period, but not its timeset or edgeset name.

Any pin can be masked. The Mask property is set to True or False. You can mask or unmask a pin by changing the value in the Properties window. You can also select a pin or group in the Burst Explorer and select **Debug>Mask**. See [Masking Pins](#).

Which pin properties are displayed depends where you select the pin, in the Burst Explorer or in the display area. If you select it in the Burst Explorer, the Properties window displays levels information. If you select a pin in the pin headers column, it is the same as selecting it in the Burst Explorer. If you select a point of a pin lane in the waveform display area, you are actually selecting one or more cycles. The associated timeset is available, so the Properties window can display timing and format information and pattern execution information, in addition to levels information. Some timing and format values can be changed for debugging. Selecting a pin, group, or list in the row headers is the same as selecting it in the Burst Explorer: you have not selected cycles, so only levels information is displayed. You can select cycles only for pins, not for pin groups or lists.

If you select a range that includes multiple pins or cycles, the Properties window displays information for the rightmost cycle and bottommost pin. If you select discontinuous pins or cycles, the last pin or cycle selected is displayed.

The following paragraphs describe the kinds of properties displayed for each selection, but does not list all properties.

If you select a pin, pin group, or pin list in the Burst Explorer, the following is displayed in the Properties window:

General	Pin name, mask state (which you can change), instrument.
Levels	The pin’s voltage levels. You can modify the levels.

If you select a cycle in the waveform display, the following is displayed in the Properties window:

General	Pin name, cycle and vector number, fail state, module name, but not the mask state. You cannot change any of this information.
Levels	The pin’s voltage levels. You can modify the levels.
Timing/Format	All time set and edge set information, plus init state and start state. You can modify the timing edges, periods, and init and start states, but not items such as data format or pin setup.

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Displaying Waveforms : Properties Window : Protocol Properties

Protocol Properties

The Properties window displays properties of the selected protocol port, function, frame, or cycle. You select ports and functions in the Burst Explorer and frames and cycles in the waveform display area.

Protocol properties are read-only and are not modifiable through the Properties window. As with pin properties, what is displayed for protocol properties depends on where you make the selection, in the Burst Explorer or the header column, or in the display area itself. If you select a protocol port in the Burst Explorer, the following is displayed in the Properties window:

General	Port name and status, enabled state, and protocol family and type.
Timeout	Timeout enabled state and value (in seconds).

If you select a protocol function in the Burst Explorer, the following is displayed in the Properties window:

General	Family and function, pin and channel name, port name.
Levels	Pin type, whether it is differential, and voltage levels.

If you select a protocol frame in the waveform display area, the following is displayed in the Properties window:

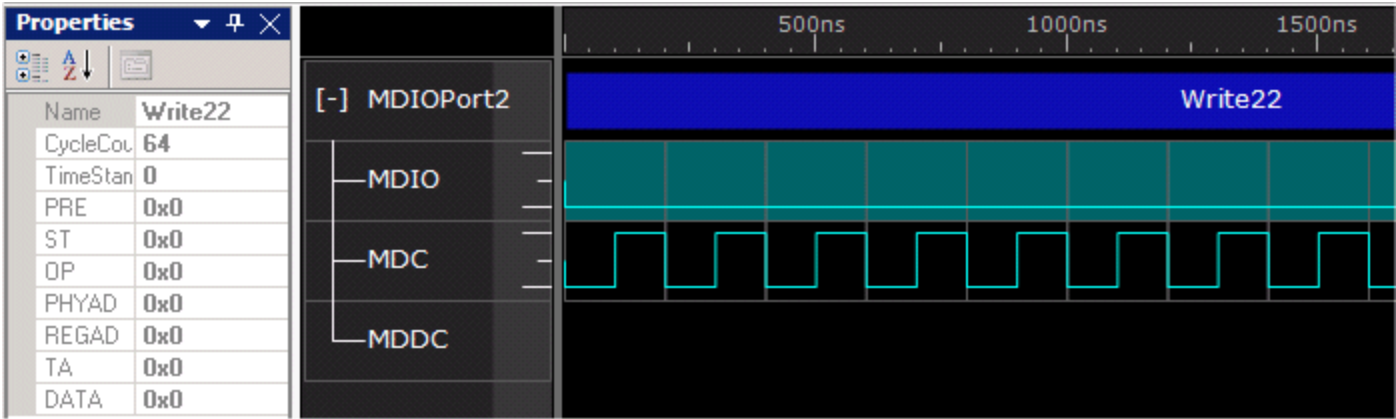
General	Name of the frame, count of cycles in the frame, timestamp of the start time of the frame.
Frame Fields	All fields in the frame and the value of each in hex.

Selecting a frame in the display area highlights all cycles in that frame that are part of a protocol-defined field.

If you select a protocol cycle in the waveform display area, the following is displayed in the Properties window:

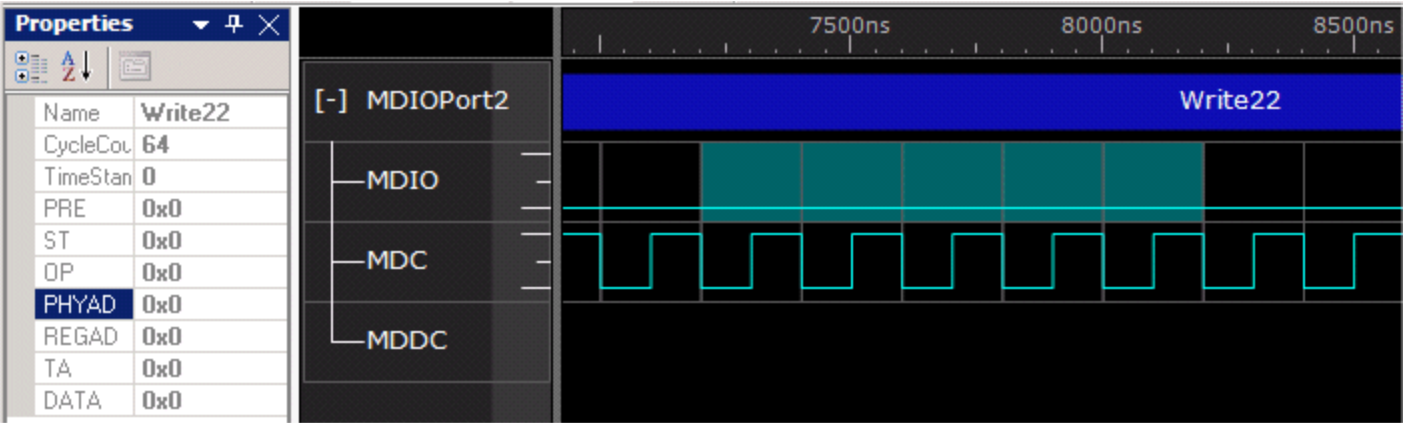
General	Cycle index, masked and failed state, bit action, and frame, function, and port name.
Levels	Pin type, whether it is differential, and voltage levels.
Timing	Period, edges.

The Properties window can provide a convenient way to navigate within frames. A frame can contain a lot of data. For example a MDIO Write22 frame contains 64 cycles. Typically, at a reasonable resolution, you cannot fit all these cycles in a single viewable area of the Waveform Display. To navigate to specific cycles within a frame, you can select the frame by clicking the port lane:



Selecting a Frame Using a Port Lane

The Properties window displays all fields that compose the frame. Select the field you want by clicking it in the Properties window. This now navigates to the cycles that make up that field:



Selecting a Field in a Frame

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WaveformDisplayDefTopic



About Waveform Display

About Waveform Display

Waveform Display provides a multi-time-domain, horizontal, waveform-view representation of pins in a pattern or in Protocol Aware (PA) frames. Supported instruments include HSD, Serial, nWire, DDR, PRBS, and Loopback.

Waveform Display can display one capture-all HRAM’s worth of cycles for HSD and Serial and one individual frame or module’s worth of cycles for PA instruments.

Using Waveform Display, you can reburst a pattern, run Actual Mode, and inspect failures and levels and timing.

You usually use Waveform Display with other debugging tools. For example, when you are debugging a digital pattern, you may start by examining the pattern in PatternTool. At some point, you may want to view specific waveforms and can navigate to Waveform Display to do it. To debug a Protocol Aware problem, you may start with Protocol Studio and then navigate to Waveform Display.

This document covers the following topics:

- | | |
|---|--|
| • | Displaying Waveforms |
| • | Bursting, Actual Mode, and HRAM |
| • | Waveform Display Toolbar and Command Reference |

IG-XL includes additional digital tools:

- | | |
|---|--------------|
| • | PatternTool: |
|---|--------------|

- Lets you create, edit, and debug pattern files

- Lets you overlay failures on programmed vectors

- Pattern Control Display:

- Displays pattern information and memory consumption

- Provides interactive pattern execution and management

- HRAM Display:

- Provides a tabular view of captured HRAM data

- Includes fail overlay and actuals

- Protocol Studio:

- Displays Protocol Aware (PA) frames at various levels of abstraction

- Enables programming of PA port settings

- Enables initiation of frames

- Protocol Definition Editor:

- Lets you define a protocol from scratch

- Lets you edit an existing protocol definition

- Lets you graphically view frame waveforms

- CyclePro:

- Provides raw cycle display mode and symbol10b display mode
- Decodes PCIe Link traffic (CMEM) and ATE Tx traffic (Tx HRAM)
- Provides layer selection: Transaction, Data Link, PHY

For information on other IG-XL digital tools, see:

- About PatternTool
- Pattern Control Display
- About HRAM Display
- About Protocol Studio
- About Protocol Definition Editor
- About CyclePro

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WaveformDisplayRef.4.1



Waveform Display Toolbar and Command Reference

Waveform Display Toolbar and Command Reference

This section lists the actions that are available from the Waveform Display menus and toolbars. References to more complete descriptions in Help are included. Most toolbar buttons correspond to menu commands, but some are available only on the toolbars. The section is organized on where the command appears, on the toolbar or on one of the menus:

- [Toolbars](#)
- [File Menu Commands](#)
- [View Menu Commands](#)
- [Debug Menu Commands](#)
- [ActualMode Menu Commands](#)
- [Tools Menu Commands](#)



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WaveformDisplayRef.4.2

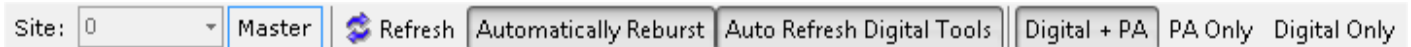


Waveform Display Toolbar and Command Reference : Toolbars

Toolbars

Waveform Display has two toolbars. Most of the buttons are also available as menu commands, but some toolbar controls appear only on the toolbar.

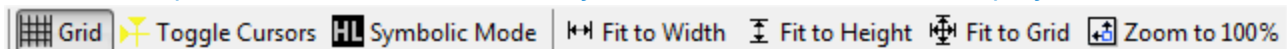
The first part of the first toolbar mainly controls what is displayed:



Waveform Display First Toolbar, First Part

	Selects the site, either using the master site as set in DataTool or selecting a site. See Selecting the Site .
	View>Refresh or F5 Refreshes the display. This is needed when the Waveform Display title bar indicates that a refresh is required. See Refreshing and Rebursting .
	Enable or disable automatically rebursting when you click Refresh or bring up the tool. See Turning Off Automatic Rebursting .
	Refreshes other open digital tools when Waveform Display performs a burst. See Refreshing Other Digital Tools .
	Displays digital (including serial) and PA data, or only digital or only PA. See Selecting Digital or PA Mode .

The second part of the first toolbar mainly controls how the data is displayed:



Waveform Display First Toolbar, Second Part

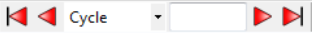
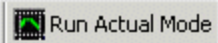
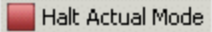
 Grid	View>Grid Toggles the grid on and off. See Using the Grid.
 Toggle Cursors	View>Cursors Toggles the cursors on and off. See Using Cursors.
 Fit to Width	Expands the selection horizontally so that it fills the width of the display area. The height is unchanged. See Using Zoom.
 Fit to Height	Expand the selection vertically so that it fills the height of the display area. The width is unchanged. See Using Zoom.
 Fit to Grid	Expands the selection so that it fills both the height and the width of the display area. See Using Zoom.
 Zoom to 100%	Restores the horizontal zoom to 100%. See Using Zoom.

The second toolbar mainly controls bursting, Actual Mode, and navigation:



Waveform Display Second Toolbar

 PrePatF	Enables or disables the execution of the PrePatF function before each pattern burst. See Burst Commands.
 PostPatF	Enables or disables the execution of the PostPatF function after each pattern burst. See Burst Commands.
 Burst	Bursts the most recently burst pattern or pattern set. The options are: - Burst - Burst Loop - Burst Until Pass - Burst Until Fail - Burst Time Align

See Burst Commands.

Halts the currently bursting pattern. This button is enabled only when a pattern is being burst. See Refreshing and Rebursting.

Navigates to the first, previous, next or last instance of the selected item. See Goto.

Runs actual mode. The button is enabled only if there is a valid actual mode setup. See Using Actual Mode.

Halts actual mode. The button is enabled only when actual mode is running. See Using Actual Mode.





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WaveformDisplayRef.4.3



Waveform Display Toolbar and Command Reference : File Menu Commands

File Menu Commands

Exit	Causes Waveform Display to close.
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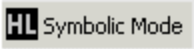
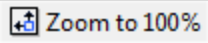
WaveformDisplayRef.4.4



Waveform Display Toolbar and Command Reference : View Menu Commands

View Menu Commands

Burst Explorer	Displays the Burst Explorer panel and gives it focus. See Burst Explorer.
Properties	Displays the Properties panel and gives it focus. See Burst Explorer.
Message Panel	Displays the Messages panel and gives it focus. See About the Message Panel.
Actual Mode Setup	Displays the Actual Mode panel and gives it focus. See Using Actual Mode.
HRAM Setup	Displays the HRAM Setup panel and gives it focus. See HRAM Setup.
Refresh	 Refresh F12 Refreshes the display to reflect the latest state of the system. See Refreshing and Rebursting.
Take Snapshot	Takes a snapshot of the display section. See Taking and Viewing Snapshots.
View Snapshot	Displays the snapshots. A pull-right menu lets you view the most recent snapshot or select a different snapshot. See Taking and Viewing Snapshots.
Grid	 Grid Toggles the grid on and off in the waveform display area. See Using the Grid.
Cursors	 Toggle Cursors

	Toggles the cursors on and off in the waveform display. See Using Cursors.
Symbolic	 Toggles between waveform and symbolic modes for the waveform display. See Using Symbolic Mode.
Zoom	Has the following zoom options: • Fit Selection to Height • Fit Selection to Width • Fit Selection to Full Grid For the meaning of the options as well as the corresponding toolbar buttons, see Using Zoom.
Restore Default Settings	Restores the tool to its initial default condition. For example, panels that were dismissed are redisplayed, the grid is turned on, cursors are turned off, and initial row heights are restored.
Zoom to 100%	 Restores the horizontal and vertical zoom to 100%. See Using Zoom.

 		
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WaveformDisplayRef.4.5

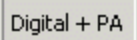
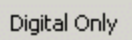
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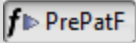
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Waveform Display Toolbar and Command Reference : Debug Menu Commands

Debug Menu Commands

These commands are related to bursting patterns. The Burst commands are available on the Burst button drop-down. For details on these operations, [Bursting Patterns](#).

Burst	 Burst	Rebursts the last burst patterns or pattern sets in the job.
Burst Loop	 Burst Loop	Rebursts in a loop. Use Halt to stop the loop.
Burst Until Pass	 Burst Until Pass	Rebursts in a loop, until the first pass on an HSD pin is detected.
Burst Until Fail	 Burst Until Fail	Rebursts in a loop, until the first failure on an HSD pin is detected.
Burst Time Align	 Burst Time Align	Rebursts a pattern with both HSD and serial pins in a way that permits correct time alignment. See Burst Time Align.
Halt	 Halt	Halts the currently the currently bursting pattern. This button is enabled only when a pattern is being burst.
Mask		Toggles the masking on the currently selected pin in the Burst Explorer. See Masking Pins.
Digital + PA PA Only	  	

Digital Only	Displays digital (including serial) and PA data, or only digital or only PA. See Selecting Digital or PA Mode.
PrePatF	 PrePatF Enables or disables the execution of the PrePatF function before each pattern burst.
PostPatF	 PostPatF Enables or disables the execution of the PostPatF function after each pattern burst.

 		
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WaveformDisplayRef.4.6



Waveform Display Toolbar and Command Reference : ActualMode Menu Commands

ActualMode Menu Commands

Run Actual Mode	 Run Actual Mode	Runs Actual Mode on the currently selected section of the displayed waveforms.
Halt	 Halt Actual Mode	Halts the running of Actual Mode.
Clear		Clears the Actual Mode traces from the selected section of the waveform display area.
Clear All		Clears the Actual Mode traces from the entire waveform display area.
Export Last Actual Mode Run		Saves the most recent Actual Mode run as a .wav file in the program directory. The .wav file can be opened in WaveScope. See Steps for Using Actual Mode .

For more information, see [Using Actual Mode](#).



WaveformDisplayRef.4.7



Waveform Display Toolbar and Command Reference : Tools Menu Commands

Tools Menu Commands

These commands bring up other digital tools or let you select nondefault settings.

Pattern Tool	Opens PatternTool to display the currently displayed pattern or pattern set.
Protocol Studio	Opens Protocol Studio to display the transactions executed during this session.
HRAM Display	Opens HRAM Display with the results from the most recent burst.
Settings	Lets you select nondefault tool settings. See Other Tool Settings.
Auto Refresh Digital Tools	Auto Refresh Digital Tools Refreshes other open digital tools when Waveform Display performs a burst. See Refreshing Other Digital Tools.
Automatically Reburst	Automatically Reburst Turns on or off automatically rebursting the pattern. See Turning Off Automatic Rebursting.



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